



A multi-objective artificial electric field algorithm with reinforcement learning for milk-run assembly line feeding and scheduling problem

Binghai Zhou^{*}, Mingda Wen

School of Mechanical Engineering, Tongji University, Shanghai 201804, China

ARTICLE INFO

Keywords:

Multi-objective artificial electric field algorithm
SARSA selection mechanism
Milk-run
Scheduling
Epsilon constraint method

ABSTRACT

In automotive mixed-model assembly lines (MMALs), a large number of different parts need to be supplied to the assembly lines on time, which poses significant logistical challenges for manufacturers. However, consistently supplying parts for MMALs is a very complex issue due to factors such as diverse component requirements and logistical coordination in the supply chain. In this paper, we propose a bi-objective optimization problem to minimize the line-side inventory and energy consumption in a milk-run material distribution system. Meanwhile, the number of Kanban and the capacity of the material bin that affect the scheduling is jointly optimized, so that the material distribution scheduling plan is optimized. Considering the character of the problem, a multi-objective artificial electric field algorithm with SARSA mechanism (MOAEFASA) is developed to solve the problem. The algorithm proposed combines the merits of the artificial electric field algorithm (AEFA) and the framework of the non-dominated sorting genetic algorithm (NSGA-II). In addition, several optimization strategies are used to optimize the performance of the algorithm. Finally, the validity of the mathematical model is verified through the Epsilon constraint method and the superiority of the MOAEFASA is illustrated by numerical experiments with four outstanding meta-heuristics.

1. Introduction

Today, automotive assembly systems are faced with formidable challenges stemming from the escalating diversity of customer demands. In order to maintain sufficient competitiveness, it has been more and more imperative for manufacturing enterprises to simultaneously enhance the flexibility and effectiveness of production lines, promoting the extensive application of MMALs (Zhou & Peng, 2019). The material distribution problem, as one of the key challenges in MMALs, takes responsibility for the timely and efficient delivery of materials to the production line, which is paramount for fulfilling Just-In-Time (JIT) production.

The challenge posed by the JIT material distribution problem lies in effectively delivering required parts to specific locations within designated time windows (Sipahioglu & Altin, 2019). The milk-run system is an important means of realizing JIT material distribution. Within this system, vehicles commence their journeys from a starting point (such as a warehouse or a line-side integrated supermarket), make stops at pre-determined locations, and ultimately return to the initial starting point. Through reasonable scheduling, milk-run systems can obviously reduce line-side inventory, hence decreasing inventory costs. Kilic et al. (2012)

propose three categories for milk-run material distribution problems. The first category is the “general assignment problem,” where both the route and time periods are unknown. The second is the “dedicated assignment problem,” wherein the route is predetermined, but the operating time of the vehicle on the specified route remains unknown. Lastly, the “determined time periods assignment problem” involves known time periods for vehicle travel, but unknown routes. At the same time, they gave a mathematical model for the determined time periods assignment problem. However, in real factories with MMALs, vehicles typically have multiple predetermined routes. The time required for a vehicle to deliver parts needs to be determined through scheduling. Employing a milk-run system enables companies to establish reliable and frequent transportation schedules, facilitating the distribution of materials to assembly lines in a JIT manner. This approach ensures a steady supply of parts, reduces the necessity for excessive inventory, and maintains uninterrupted production. Consequently, exploring solutions to the multi-route dedicated assignment problem holds significant practical value.

However, with the growing emphasis on sustainable production models and energy-related concerns, solely optimizing inventory costs is

^{*} Corresponding author.

E-mail addresses: bhzhou@tongji.edu.cn (B. Zhou), 2130229@tongji.edu.cn (M. Wen).

<https://doi.org/10.1016/j.cie.2024.110080>

Received 6 September 2023; Received in revised form 20 December 2023; Accepted 14 March 2024

Available online 16 March 2024

0360-8352/© 2024 Elsevier Ltd. All rights reserved.

no longer the sole focus of the automotive industry (Briand et al., 2018). Instead, the consideration of energy costs in transportation is drawing a growing concern. However, MMALs following the JIT principle typically distribute materials in small batches, which conflicts with the goal of energy conservation. When vehicles are operated with higher frequency, energy consumption in transportation increases while inventory costs decrease, and vice versa. Therefore, it is of strategic significance and practical value to incorporate energy factors when studying the scheduling problem of MMALs using the milk-run approach. Based on these considerations, the primary motivation of this paper is to concurrently optimize the economic and energy-efficient performance of milk-run material distribution systems in MMALs.

In this paper, a milk-run material distribution scheduling problem considering energy factors (MMDSPCE) is proposed for MMALs. The material distribution schedule for heterogeneous AGVs with defined travel routes is developed. Moreover, the Kanban quantity and the capacity of material bins that affect the scheduling plan are jointly optimized to realize the optimization of the material distribution schedule. When formulating the material distribution schedule, the two objectives of minimizing total line-side inventory (TLI) and total energy consumption (TEC) are considered together, which correspond to the JIT principle and energy efficiency standards respectively. Due to the non-deterministic polynomial (NP-hard) nature of the problem, a multi-objective artificial electric field algorithm with SARSA selection mechanism (MOAEFASA) is developed to generate the optimal dispatch plan of the problem by considering the delivery route, delivery time, and material loading. MOAEFASA is a hybrid algorithm of AEFA and NSGA-II, in which fast non-dominated sort and crowding distance are applied to select offspring. Moreover, the SARSA selection mechanism is introduced to select the optimal local search scheme and improve the local search ability of the algorithm. Besides, a diagonal linear random initialization strategy is proposed in the phase of population initialization to increase the diversity of the population. Finally, experimental results show that the proposed MOAEFASA outperforms the comparative algorithms, and the obtained solutions have better Pareto fronts.

This paper makes the following four contributions:

- (1) The MMDSPCE for MMALs is proposed. To meet the requirements of JIT delivery and energy saving together, the energy consumption target is incorporated into the milk-run material distribution problem. Consequently, a bi-objective problem is formulated, aiming to simultaneously minimize TLI and TEC. Additionally, a joint optimization is conducted for the Kanban quantity and the material bin capacity, both of which impact the efficiency of the milk-run system.
- (2) A mathematical model for the optimization of the proposed MMDSPCE is established. Moreover, the accuracy of the model is demonstrated with the help of the Epsilon constraint method.
- (3) Considering the complexity of the problem, this paper proposes a multi-objective intelligent optimization algorithm, MOAEFASA, to solve the MMDSPCE. To improve population diversity and local search capability of the algorithm, several optimization strategies are proposed.
- (4) The validity and superiority of the algorithm are verified through the comparison experiment with four multi-objective optimization algorithms. Additionally, a specific instance is employed to exemplify managerial applications, illustrating the practicality and significance of the proposed model and solution method.

The remainder of the study is organized as follows: Section 2 presents the literature review. Section 3 describes the problem and introduces the mathematical model and related property theorems. Section 4 develops the MOAEFASA algorithm. Section 5 conducts computational experiments to evaluate and verify the performance of MOAEFASA. Finally, the study is summarized in Section 6 and an outlook for future research is presented.

2. Literature review

In recent years, due to the gradual increase of product customization and renewal requirements, the research on the material distribution scheduling problem of MMALs has become more and more urgent, and the material distribution system, as one of the most vital parts, has attracted extensive attention from scholars (Emde & Schneider, 2018; Hou et al., 2023). Furthermore, the milk-run distribution approach, as demonstrated by Domingo et al. (2007), can successfully standardize the material distribution process, making it easy to manage and control in practical implementation. Up till now, the application of milk-run distribution systems in plants has been thoroughly investigated. Kilic et al. (2012) systematically partitioned the problem based on distribution routing and cycle time. Then, with the goal of minimizing the fixed and variable costs of vehicles, they developed the mathematical models of the material distribution scheduling problem with fixed distribution times on a single route and multiple routes. Moreover, Ciernocizowski and Bozer (2013) pointed out that in a stable milk-run distribution system, the number of Kanbans and the physical volume of vehicles have a greater impact on the shortage of workstations, so they studied the calculation method of the number of Kanbans in the milk-run distribution systems. However, their research is limited to the single-route and single-vehicle milk-run distribution systems. Further, Faccio et al. (2013) studied how to apply Kanban and optimize Kanban quantities in multi-model assembly lines. In this study, the product BOM and production mix variations are considered. In addition, Hanson and Finnsgrård (2014) studied the effect of material bin capacity on material supply efficiency, but they only conducted a qualitative study. Sipahioglu and Altin (2019) established an integer programming model for an in-plant milk-run distribution system to determine automatic guided vehicle (AGV) distribution routes and cycle time. Their proposed model allows for split deliveries at the workstations and considers increasing AGV capacity by adding trailers. Besides, an excel-based method for optimizing milk-run distribution systems was designed by Francuz and Bányai (2022). They determined a complete distribution route by establishing routes between delivery objects. Zhou and Zhao (2022) fully considered the impact of fixed usage cost and variable operation cost of EVs on production cost. They developed a mathematical model for scheduling EVs in the milk-run material supply system, aligning with energy-saving objectives by minimizing the quantity of EVs and their maximum driving distance. However, their study focused on the EV handling process without considering the amount of line-side inventory. Miqueo et al. (2023) developed a discrete event simulation model based on the milk-run distribution model using an industrial case study of a global white goods manufacturing company. They simulated four scenarios and presented the simulation results. Finally, Facchini et al. (2023) investigated the practical integration of a milk-run system with an internet of things system, proposing a dynamic routing strategy for the planning of towing trains. An algorithm developed is proposed and applied to a real case study of a global automotive company. Summarizing the above literature findings, the core issue that needs to be considered in a milk-run distribution system is determining the material distribution route and cycle time. However, factors such as Kanban quantity, material bin capacity, handling vehicle capacity, material distribution method and the introduction of new systems or technologies all have great impacts on the scheduling plans formulated. In particular, Kanban quantity and material bin capacity will have a direct impact on how stable the assembly system is, thus considering their combined optimization is crucial to maintaining the stability of the assembly system and reducing the line-side inventory. However, there is currently very little literature that considers the influence of these two factors together in the milk-run distribution system.

From another perspective, the Kanban quantity and the material bin capacity also have impacts on the frequency of material distribution, however, higher distribution frequency also means increased energy consumption. How to achieve the trade-off between the two is an urgent

challenge that needs to be addressed today. Nowadays, with increasingly stringent government requirements for sustainable development, more and more scholars have conducted research on reducing the energy consumption of discrete manufacturing systems (Diaz & Ocampo-Martinez, 2019). For instance, Zhou and Shen (2018) proposed a material distribution scheduling problem considering energy consumption. They designed an automatic material handling system that uses a group of AGVs to complete the material distribution tasks. Finally, under the condition of ensuring the stability of the assembly systems, the control of the energy consumption of the material distribution process is realized. Barak et al. (2021) studied the energy consumption of AGVs in resource-constrained flexible manufacturing systems. A linear energy consumption function based on load and distance is established. Zhou and Zhu (2021) used mobile robots in the material distribution process. They took the total energy consumption of mobile robots as one of the goals, and hoped to design an energy-saving scheduling system. Vahedi-Nouri et al. (2022) discussed a bi-objective cooperative capacity-constrained electric vehicle routing problem, aiming to minimize total cost and total electrical energy consumption. Bányai (2023) investigated the application of drones for in-plant transportation of small parts. They proposed a joint AGV-Drone in-plant supply model, which makes possible the joint service of AGV-based milk-run vehicles and drones for in-plant supply. Akkad and Bányai (2023) described how to transform the traditional in-plant supply solution based on milk-run operation to a physical networked milk-run operation supply. An approach to in-plant supply systems based on industry 4.0 technology is proposed. A mathematical model with the goal of minimizing energy consumption and carbon emissions is given. The proposed approach involves the introduction of manufacturing execution system and the application of industry 4.0 technologies such as digital twins, which have good research and application prospects. In addition, the relationship between the energy consumption of AGVs and factors such as load, distance, time, and acceleration have all been considered, and the energy consumption models for which were also established respectively (Briand et al., 2018; Zhang et al., 2014). To sum up, while paying attention to the stability of the assembly systems, research on the energy consumption index of the distribution process is equally important.

In terms of the solution method, since the proposed MMDSPCE is a classical NP-hard problem, it will take a lot of time to solve using the exact algorithm when the problem scale is large (Boysen & Emde, 2014; Emde et al., 2018; Kilic & Durmusoglu, 2013; Kilic et al., 2012). In recent decades, meta-heuristic algorithms have been used more often to solve complex optimization problems due to their excellent computational efficiency. Therefore, many well-known meta-heuristic algorithms have been proposed and widely used to solve NP-hard problems, such as genetic algorithm (GA) (Tasoglu & Ali Ilgin, 2023), particle swarm optimization (PSO) (Wang et al., 2023). In addition, the AEFA is a swarm intelligence algorithm that simulates the motion of charged particles in an electrostatic field. This algorithm was originally proposed by Anita and Yadav (2019), whose main features include few optimization parameters, fast convergence, and high accuracy (Cao et al., 2022). Subsequently, Anita et al. (2020) extended the AEFA for constrained optimization problems by introducing new velocity and position constraints based on the original AEFA. Izci et al. (2020) proposed an artificial electric field with Nelder-Mead algorithm, which combines AEFA with the simplex method. They compared the proposed algorithm with the original AEFA and several improved AEFA to verify the effectiveness of the proposed algorithm. Besides, Bi et al. (2022) proposed an AEFA with inertia and repulsion, which introduced an inertia mechanism and mutual repulsion between charges based on the original AEFA. They applied the proposed algorithm to solve the spherical mining spanning tree problem and the results proved its superiority. Moreover, Niroomand (2021) applied the improved AEFA to solve the linear assembly line balance problem for a single type of product. They use the classical simulated annealing algorithm as a local search mechanism, which effectively improves the local search ability.

Adegboye and Deniz Ulker (2023) proposed a hybrid AEFA using cuckoo search algorithm and refraction learning. The cuckoo search method and refraction learning are used to improve the global exploration capability and local exploitation potential of the underlying AEFA algorithm.

Furthermore, since the proposed problem is a bi-objective optimization problem, the basic single-objective optimization algorithms cannot meet the computational demands. Since Coello (2006) first introduced evolutionary algorithms to solve multi-objective problems, many classic multi-objective optimization algorithms have been proposed to solve practical engineering problems (Elmi et al., 2023; Hou et al., 2023). In recent years, some new multi-objective optimization algorithms and their variants have been shown to have superior performance. Peng and Zhou (2019) proposed a hybrid bi-objective gray wolf optimization algorithm that combines a decomposition framework and a gray wolf optimizer when solving a bi-objective truck scheduling problem. Petwal and Rani (2020) proposed a MOAEFA. They utilized Pareto strength for fitness assignment and adopted a fine-grained elite selection mechanism to maintain population diversity. Furthermore, Naderipour et al. (2021) used the MOAEFA to configure wind turbines in power distribution systems. They aimed to reduce power losses and voltage deviations simultaneously. The numerical experiments showed that it outperforms MOGWO and MOPSO. In addition, Premkumar et al. (2022) proposed a multi-objective equilibrium optimizer for complex optimization problems, which combines a non-dominated sorting strategy and a crowding distance mechanism to ensure population diversity. After verification on 33 problems, the results demonstrate that the proposed algorithm performs well in terms of solution efficiency, robustness and exploration ability. More innovatively, Liu et al. (2022) proposed an adaptive multi-objective differential evolution algorithm based on the SARSA method. They applied the SARSA mechanism to adaptively select appropriate constraint processing techniques and generate strategies. Experimental results also showed that it has the best overall performance in solving specific types of multi-objective optimization problems. Zhou and Zhao (2023) proposed an adaptive artificial bee colony algorithm enhanced by deep Q-learning network (AABC-DQN) in solving the milk-run vehicle scheduling problem based on the supply hub model considering parts rationing. To extend the search scope, they propose to utilize deep Q-learning network to select the domain search operator.

However, according to the No Free Lunch (NFL) theorem (Wolpert & Macready, 1997), no single algorithm is suitable for any scheduling problem. Each algorithm has its advantages and limitations. Therefore, in this study, an efficient hybrid algorithm is proposed to solve the MMDSPCE. First, a diagonal linear random initialization strategy is used to generate the initial population. Then, four local search operators are proposed, and the SARSA mechanism is introduced for selection, which effectively improves the local search ability of the algorithm. Finally, two repair mechanisms are designed to fix infeasible solutions.

3. Problem description and mathematical model

3.1. Problem description

Fig. 1 shows the milk-run material distribution system for MMALs. The overall operation process of the system is as follows. In terms of assembly line inventory, for the production cycle $t \in T$, when a bin of part $i_0 \in I_s$ is consumed in workstation $s \in S$, which corresponds to $NK_{i_0}^t \leq NK_{i_0} - 1$, workstation s generates a replenishment Kanban for part i_0 . The earliest replenishment time $t_{s i_0}^{min}$ is obtained. For production cycle t , when the line-side inventory quantity $I_{i_0}^{ts}$ of part i_0 minus the production demand $d_{i_0}^{ts}$ of the current production cycle fails to satisfy the production demand $d_{i_0}^{t+1,s}$ of the next production cycle, which corresponds to $I_{i_0}^{ts} - d_{i_0}^{ts} \leq d_{i_0}^{t+1,s}$, workstation s will issue a mandatory replenishment demand for part i_0 to obtain the latest replenishment time

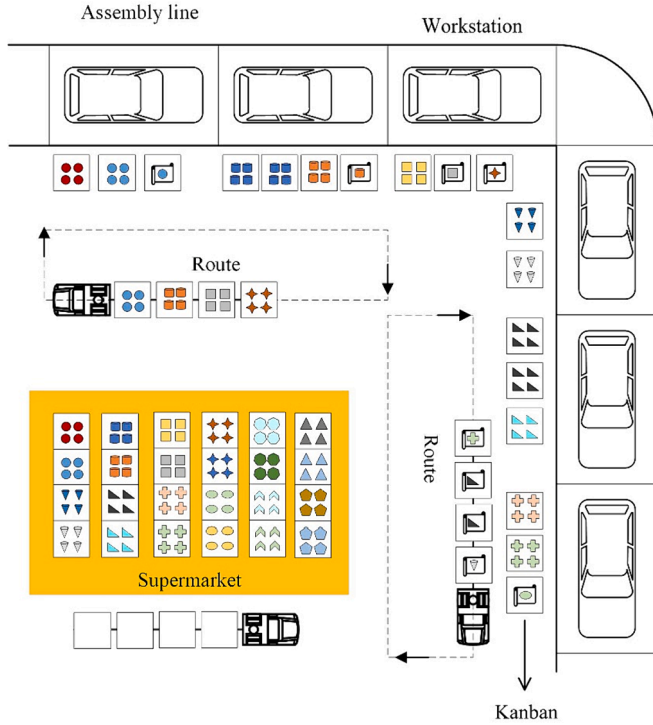


Fig. 1. Milk-run material distribution system for mixed-model assembly line.

t_{slo}^{max} . Part i_0 needs to be replenished within the replenishment time window $T_{slo}^{rep} \in [t_{slo}^{min}, t_{slo}^{max}]$. In terms of transportation, for production cycle t , the multi-load AGV $v \in V$ obtains distribution route $r \in R$ based on a route scheduling decision z_{vr}^t . Based on the load scheduling decision z_{ir}^{st} on the route r , AGV v obtains the parts i to be delivered. The AGV v loads the parts to be delivered at the material supermarket with a loading time of tl per bin. Then AGV v passes by workstation $s \in S_r$ on route r in turn, and replenishes a bin for part i_0 that needs to be restocked, which $I_{i_0}^{st} + Cb_{i_0}$. The unloading time for each bin is tr . At the same time, AGV v gets information about the replenishment Kanbans of the workstations. When the replenishment tasks for all workstations on route r are completed, the AGV v returns to the material supermarket and waits for new allocation tasks for the next production cycle. The notation definitions of all parameters are given in Table 1.

Based on the problem description above, the Kanban quantity NK_i and the material bin capacity Cb_i of part i can have a significant impact on the scheduling plan developed. Therefore, optimizing the Kanban quantity and the material bin capacity for each part is considered in this paper to achieve the optimization of the material distribution system. In general, on the one hand, the higher the Kanban quantity of a part, the higher its line-side inventory. While on the other hand, when the maximum actual loading capacity of a bin is reduced, the amount of line-side inventory is effectively reduced with the same Kanban quantity. More specifically, when parts have higher Kanban quantities and lower bin capacities, material is delivered more frequently and therefore consumes significantly more energy, but the number of line-side inventories stays low. However, when parts have lower Kanban quantities and higher bin capacities, the number of line-side inventories is higher, but the number of part replenishment conditions triggered is significantly lower, so less energy is consumed. In summary, the MMDSPCE studied in this paper establishes and optimizes the material distribution scheduling scheme by setting the optimal Kanban quantity and material bin capacity for each part, achieving the goal of reducing line-side inventory and reducing energy consumption in the distribution process.

Table 1
Parameter definition.

Notations	Definitions
Sets	
T	Set of production cycles, $t \in T$
S	Set of workstations, $s \in S$. When $s = 0$, it means material supermarket.
V	Set of AGVs, $v \in V$
I	Set of parts, $i \in I$
R	Set of routes, $r \in R$
Parameters	
I_i	Set of parts used by workstation s
S_r	Set of workstations that route r contains
d_i^{st}	Number of parts i required by workstation s in production cycle t
dis_s	Distance between the material supermarket and workstation s
dis_r	Distance traveled by route r
CT	Duration of a production cycle
Cl	Maximum line-side inventory capacity of the workstations
Cb_{min}	Minimum capacity of material bin
Cb_{max}	Maximum capacity of material bin
NK_{min}	Minimum Kanban quantity of parts
NK_{max}	Maximum Kanban quantity of parts
Cv	Maximum load capacity of an AGV
θ	Minimum load rate for an AGV
we_i	Weight of a bin of part i
we_v	Weight of an empty AGV
tl	Time to load a bin of parts into an AGV
tr	Time to unload a bin of parts from an AGV
w	Speed of the AGV
Variables	
T_{si}^{rep}	Production cycle set of replenishable parts i used in workstation s
t_{si}^{min}	Earliest production cycle for workstation s to reorder part i
t_{si}^{max}	Latest production cycle for workstation s to reorder part i
$I_{i_0}^{st}$	Inventory of part i stored at workstation s during production cycle t
t_{si}^{rep}	Replenishment time of part i used in workstation s
we_r^t	Weight of parts distributed by the AGV for workstation s on the route r in production cycle t
NK_i^t	Bin quantity of part i stored at the assembly line in production cycle t
Decision variables	
z_{vr}^t	Binary variable whose value is 1 when the AGV distributes parts for route r in production cycle t and 0 otherwise
z_{ir}^{st}	Binary variable whose value is 1 when the AGV distributes part i for workstation s on route r in production cycle t and 0 otherwise
NK_i	Kanban quantity of part i
Cb_i	Quantity of part i loaded in a material bin

3.2. Problem assumption

To formulate the problem effectively, the following assumptions are proposed.

- (1) The production plan and material requirements of each workstation are determined in advance.
- (2) The cycle time is taken as the basic time unit of the assembly line.
- (3) The AGVs and the border of the line for workstations have capacity constraints.
- (4) The loading and unloading times of a material bin are known and fixed.
- (5) Alternative material distribution routes are known and fixed.
- (6) After the completion of one material distribution task, AGVs shall wait until the beginning of the next production cycle to execute the next material distribution task.
- (7) The AGVs can distribute materials only when the minimum load requirements are met.
- (8) The AGVs travel at a constant speed.

3.3. Mathematical model

Table 1 lists the notations and definitions for all the parameters.

First, because each part on the assembly line has different Kanban quantity and material bin capacity, each part has a different time window for replenishment. The replenishment time window T_{si}^{rep} for parts is

defined as follows.

$$T_{si}^{rep} = (t_{si}^{min}, t_{si}^{max}) \quad (1)$$

Where, t_{si}^{min} represents the earliest replenishment time of part i , which can be calculated by Eq. (2), t_{si}^{max} represents the latest replenishment time of part i , which can be calculated by Eq. (3).

$$NK_i^t \leq NK_i - 1, \quad i \in I, t \in T \quad (2)$$

$$I_v^{ts} - d_i^{ts} \leq d_i^{t+1,s}, \quad i \in I_s, s \in S, v \in V \quad (3)$$

Then, we define the energy consumption model of multi-load AGV. In this paper, the energy consumption model proposed by Briand et al. (2018) is reasonably simplified by taking no account of the acceleration and deceleration process of AGVs. Therefore, the simplified energy consumption model is as follows.

$$E = \int \gamma \hat{A} \cdot m(t) \bullet v(t) dt \quad (4)$$

Where, γ represents energy consumption rate, $m(t)$ and $v(t)$ represent the functions of the mass and velocity of the vehicles with time.

Finally, according to the problem assumptions and parameter notations, the mathematical model of the problem is as follows, which is completed based on three researches (Kilic et al., 2012; Sipahioglu & Altin, 2019; Zhou & He, 2021).

$$Minimize f = \{f_1, f_2\} \quad (5)$$

$$f_1 = \sum_{i \in T} \sum_{s \in S} \sum_{i \in I} I_v^{st} \quad (6)$$

$$f_2 = \sum_{i \in T} \sum_{v \in V} \sum_{r \in R} \sum_{s \in S_r} \gamma z_{vr}^{st} (w e_v + w e_r^{st}) dis_s \quad (7)$$

$$I_v^{st+1} = I_v^{st} - d_i^{st} + z_{ir}^{st} C b_i, \quad i \in I_s, s \in S, t \in T \quad (8)$$

$$w e_r^{st} = \sum_{i \in I_s} z_{ir}^{st} w e_i, \quad s \in S_r, r \in R, t \in T \quad (9)$$

$$\sum_{v \in V} z_{vr}^{st} \leq 1, \quad r \in R, t \in T \quad (10)$$

$$\sum_{r \in R} z_{vr}^{st} \leq 1, \quad v \in V, t \in T \quad (11)$$

$$\sum_{s \in S_r} \sum_{i \in I_s} z_{ir}^{st} \bullet (tl + tr) + z_{vr}^{st} dis_r / w \leq n_{vt} CT, \quad v \in V, r \in R, t \in T \quad (12)$$

$$I_v^{st} - d_i^{st} \geq 0, \quad i \in I_s, s \in S, t \in T \quad (13)$$

$$t_{si}^{min} \leq t_{si}^{rep} \leq t_{si}^{max}, \quad i \in I, s \in S \quad (14)$$

$$z_{vr}^{st} C v \geq \sum_{s \in S_r} \sum_{i \in I_s} z_{ir}^{st}, \quad v \in V, r \in R, t \in T \quad (15)$$

$$\theta z_{vr}^{st} C v \leq \sum_{s \in S_r} \sum_{i \in I_s} z_{ir}^{st}, \quad v \in V, r \in R, t \in T \quad (16)$$

$$NK_{min} \leq NK_i \leq NK_{max}, \quad i \in I \quad (17)$$

$$C b_{min} \leq C b_i \leq C b_{max}, \quad i \in I \quad (18)$$

$$\sum_{i \in I_s} NK_i \leq C l, \quad s \in S \quad (19)$$

$$z_{vr}^{st}, z_{ir}^{st} \in \{0, 1\}, \quad i \in I, s \in S, v \in V, r \in R, t \in T \quad (20)$$

In the model, the objective functions (5)-(7) contain two optimization objectives, which are minimizing the TLI on the assembly lines and

the TEC of AGVs. Constraint (8) calculates the variation of line-side inventory with production cycle time. Constraint (9) calculates the weight of the parts distributed for the workstations. Constraint (10) indicates that in each production cycle, at most one AGV distributes materials in each route. Constraint (11) ensures that in each production cycle, each AGV distributes parts for at most one route. Constraint (12) determines the time for an AGV to perform a material distribution task. Since this paper assumes that AGV distributes the material at the beginning of the production cycle, n_{vt} is a minimum integer representing the interval between the two material distribution tasks performed by AGV v during the production cycle t . Constraint (13) guaranteed that there will be no shortage of material on the assembly lines. Constraint (14) represents the replenishment time window. Constraint (15) indicates that only assigned vehicles can perform material distribution tasks. Meanwhile, the vehicle capacity limit is restricted. Constraint (16) obtains the load of the AGV to meet the minimum load requirements. Constraint (17) limits the Kanban quantity for each part. Constraint (18) restricts the material bin capacity for each part. Constraint (19) represents the capacity of the border of the lines. Constraint (20) defines binary variables.

3.4. Domain properties

To enlighten the mechanism for generating the solution, we analyze the proposed solution by means of incomplete evaluation. Therefore, the following properties and theorems are presented.

Property 1: If $d_i^{st}, C b_i, C v, I_s, S_r, T$ have been given, then the lower bound of the number of materials to be distributed by route r is $MR_r^{min} = \left\lceil \left\lfloor \left[\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right] / C v \right\rfloor \right\rceil$, where $\lfloor \bullet \rfloor$ means round up.

Proof: When $t \in T$, the total part consumption of workstation on route r can be expressed as $d_r^{sum} = \sum_{t \in T} \sum_{s \in S_r} \sum_{i \in I_s} d_i^{st}$. The number of bins consumed by the workstation on route r is $R b_r = \left\lceil \sum_{t \in T} \sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right\rceil$. When the AGV is fully loaded each time, the AGV travels the least times on route r . Therefore, $MR_r^{min} = R b_r / C v = \left\lceil \left\lfloor \left[\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right] / C v \right\rfloor \right\rceil$. Property 1 is proved.

Theorem 1: If $d_i^{st}, C b_i, C v, tl, tr, dis_r, w, CT, I_s, S_r, T, R$ have been given, without considering the replenishment time window, the lower bound of AGV quantity is $N_v^{min} = \sum_{r \in R} \left\lceil \left\lfloor \left[\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right] / C v \right\rfloor / \left\lceil |T| / \left[(C v(tl + tr) + dis_r / w) / CT \right] \right\rceil \right\rceil$, where $\lceil \bullet \rceil$ means round down, dis_r^{min} is the minimum value of dis_r , which is the shortest route.

Proof: According to property 1, the number of times that route r needs to be distributed material is at least $MR_r^{min} = \left\lceil \left\lfloor \left[\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right] / C v \right\rfloor \right\rceil$. Therefore, the total number of materials distributed of all routes is at least $MR_R^{min} = \sum_{r \in R} MR_r^{min} = \sum_{r \in R} \left\lceil \left\lfloor \left[\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right] / C v \right\rfloor \right\rceil$. In this case, since AGVs distribute materials in the form of full load, according to constraint (12), the interval time between two trips of AGVs in route r to distribute materials is $nk_r = \lfloor (C v(tl + tr) + dis_r / w) / CT \rfloor$, where $nk_r^{min} = \min\{nk_1, nk_2, \dots, nk_R\}$. However, when AGV v distributes materials with the shortest interval time, it needs to perform the most material distribution time correspondingly. Thus, the material handling time can be expressed as $MR_v^{max} = \left\lceil |T| / nk_r^{min} \right\rceil = \left\lceil |T| / \left[(C v(tl + tr) + dis_r^{min} / w) / CT \right] \right\rceil$. Therefore, the lower bound of AGV quantity can be expressed as $N_v^{min} = \frac{MR_R^{min}}{MR_v^{max}} = \sum_{r \in R} \left\lceil \left\lfloor \left[\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} / C b_i \right] / C v \right\rfloor / \left\lceil |T| / \left[(C v(tl + tr) + dis_r^{min} / w) / CT \right] \right\rceil \right\rceil$. Theorem 1 is proved.

Theorem 2: If $d_i^{st}, C b_i, tl, tr, dis_r, w, CT, I_s, S_r, T$ have been given, the theoretical lower bound of Kanban quantity of part i is $N_k_i^{min} =$

$$\left| \sum_{t \in T} d_i^{st} / (|T| \bullet Cb_i) \bullet \left| (\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} (tl + tr) / Cb_i + dis_r / w) / CT \right| \right|.$$

Proof: Assuming that part i is consumed at a uniform rate, the average consumption rate of part i is $\tau_i = \sum_{t \in T} d_i^{st} / |T|$. When $t \in T$, the total demand of part i is fixed. Therefore, when Cb_i is the maximum, the Kanban quantity of part i is the minimum. Then, the average consumption rate of the material bin loaded with part i is $\tau b_i = \sum_{t \in T} d_i^{st} / (|T| \bullet Cb_i)$. When the interval time between two replenishments for part i is the shortest, the quantity of replenishment for each time is the least, then the Kanban quantity of part i is the least. According to constraint (12), the minimum interval time between two replenishments for part i is $nk_i = \left\lfloor (\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} (tl + tr) / Cb_i + dis_r / w) / CT \right\rfloor$. Therefore, the lower bound of Kanban quantity of part i is $Nk_i^{min} = \lfloor \tau b_i \bullet nk_i \rfloor = \left\lfloor \sum_{t \in T} d_i^{st} / (|T| \bullet Cb_i) \bullet \left\lfloor (\sum_{s \in S_r} \sum_{i \in I_s} d_i^{st} (tl + tr) / Cb_i + dis_r / w) / CT \right\rfloor \right\rfloor$. Theorem 2 is proved.

4. The proposed MOAEFASA algorithm

The milk-run material distribution problem with single objective has been proven to be an NP-hard problem. This paper delves into the bi-objective milk-run material distribution problem. Building upon this foundation, we also take into account the influence of Kanban quantity

and material bin capacity. Consequently, the proposed problem has a larger solution space, which makes it difficult to find the optimal Pareto solution using the exact algorithm in finite time. Therefore, a multi-objective artificial electric field algorithm with SARSA selection mechanism (MOAEFASA) is proposed to solve the problem. [Section 4.1](#) describes the overall framework of the algorithm. [Sections 4.2–4.7](#) discuss the entire content of the algorithm.

4.1. Framework of MOAEFASA

Fig. 2 shows the algorithm framework of MOAEFASA. First, the diagonally linear random strategy is used to quickly generate a series of well-diversified initial populations. Then, according to the physical characteristics of charged particles moving in the electric field, the position change of each individual in the solution space is realized. Next, four local search operators are proposed, and the SARSA selection mechanism is introduced to select the best local optimization scheme to improve the local search ability of the algorithm. Finally, according to the problem constraints and coding principles, two repair mechanisms are proposed to repair the infeasible solutions. In addition, the selection strategy of NSGA-II is introduced. By calculating the Pareto level of all solutions and the crowding distance at the same level, the pre-NP individuals are retained as the new offspring. When the number of iterations reaches the maximum, the algorithm ends and outputs the Pareto

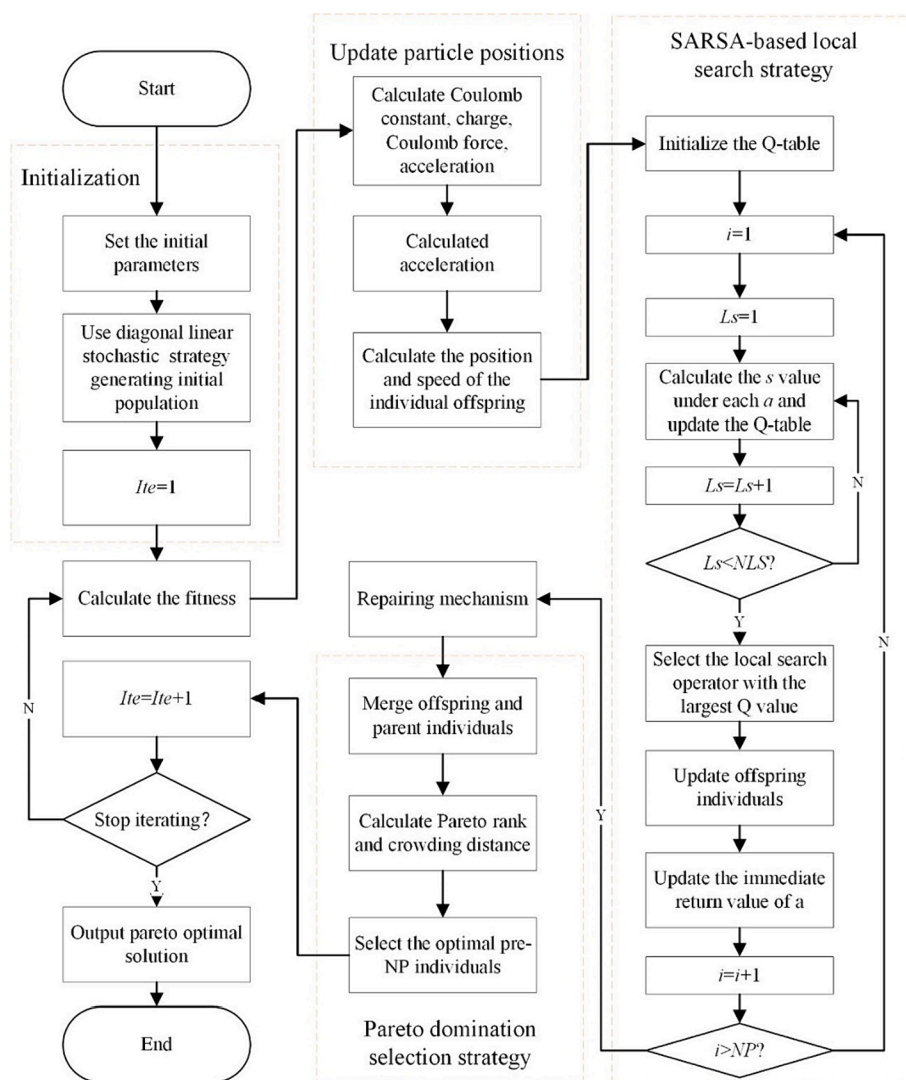


Fig. 2. Flowchart of the MOAEFASA.

optimal solution set.

4.2. Encoding and decoding

This paper needs to determine the optimal Kanban quantity and material bin capacity for each part to optimize the material distribution scheduling scheme. Therefore, the Kanban quantity and material bin capacity for each part need to be determined first. Then, for each production cycle, the types and quantities of parts that need to be distributed to the assembly lines should be identified. In order to facilitate the solution, a double-layer coding method is used to encode the problem.

As shown in Fig. 3(a) for the matrix with real number code of $2|I_s|$ in the first-layer coding of workstation s , it represents the corresponding relationship between part $i_{j,s}$, Kanban quantity $nk_{j,s}$ and material bin capacity $cb_{j,s}$. $i_{j,s}$ represents the serial number of part j on workstation s . $nk_{j,s}$ represents the Kanban quantity of part j on workstation s , which is a random integer between $[nk_{j,s}^{min}, nk_{j,s}^{max}]$. $nk_{j,s}^{min}$ can be calculated by the theorem 2, and $nk_{j,s}^{max}$ is limited by constraints (19). $cb_{j,s}$ represents the material bin capacity of part j on workstation s , which is a random integer between $[cb^{min}, cb^{max}]$. By integrating the coding from each workstation, the complete first-layer coding can be obtained.

As shown in Fig. 3(b), the length of the second-layer coding of real number coding for route r_t is $1 + |I_r|$, where $I_r = \sum_{s \in S_r} \sum_{i \in I_s} i$ represents the set of all the parts used by the workstations on route r . So, for different route r , the coding length are not exactly the same. But when production cycle t is different, the coding length of the same route is exactly the same. The second-layer coding determines the type and quantity of parts to be distributed to route r by AGVs during production cycle t . $Rv_{r,t}$ is a random real number between $[0, 1]$, which can determine whether AGVs distribute parts for route r in production cycle t by descending sorting. Besides, $i_{j,r}$ is the serial number of part j in all parts on route r . $Ri_{i,r}$ is the random real number between $[0, 1]$, and the type of parts distributed by AGVs can be determined by descending sorting. The quantity of distributed parts $Ni_{j,r}$ is determined by the demand and the line-side inventory. When $Iv_{j,t}^{st} = nk_{j,s}$, in production cycle t , part j is not needed to be distributed, then let $Ri_{j,r}^t = 0$, otherwise $Ni_{j,r} = 1$. Therefore, the first-layer coding will affect the second-layer coding. Meanwhile, according to constraint (13), when part i must be distributed for route r , let $Rv_{r,t} = 1$ and $Ri_{i,r}^t = 1$ to ensure that the assembly lines have sufficient inventory. The complete second-layer coding can be obtained by integrating the coding of each route in all production cycles. Finally, the complete scheduling plan of AGV can be obtained by integrating the first-layer coding and the second-layer coding.

In order to help understand the encoding and decoding process, the following examples are given. The number of routes is $|R| = 2$. Route r_1 through workstation s_1 and s_2 , route r_2 through workstation s_3 and s_4 . In production cycle t , the number of available AGV is 1 and the maximum capacity of AGV is 4. Part $i \in \{i_1, i_2, i_3, i_4\}$ are used in workstation s_1 . Part $i \in \{i_5, i_6, i_7\}$ are used in workstation s_2 . Part $i \in \{i_8, i_9, i_{10}\}$ are used in workstation s_3 . Part $i \in \{i_{11}, i_{12}, i_{13}, i_{14}\}$ are used in workstation s_4 . $nk_{j,s}^{min} = 2$, $nk_{j,s}^{max} = 4$, $cb^{min} = 1$, $cb^{max} = 5$. Fig. 4 illustrates the encoding and decoding process. The actual inventory and material requirements of each part are shown in Table 2. First, $nk_{j,s}$ and $cb_{j,s}$ generate random

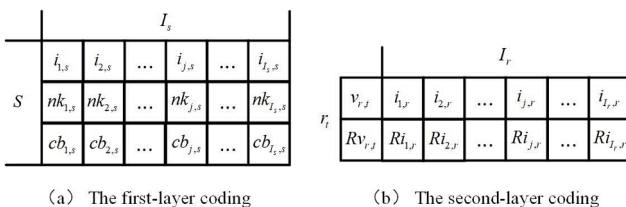


Fig. 3. The double-layer coding.

integers according to the interval $[nk_{j,s}^{min}, nk_{j,s}^{max}]$ and $[cb^{min}, cb^{max}]$, and $Rv_{r,t}$ and $Ri_{i,r}^t$ generate random numbers according to the interval $[0, 1]$. Then adjust $Rv_{r,t}$ and $Ri_{i,r}^t$ according to $nk_{j,s}$, $cb_{j,s}$, $Iv_{j,t}^{st}$ and $Iv_{j,t}^{st}$, where $Iv_{j,t}^{st}$ represents the number of bins of part j actually stored at the border of the lines. Finally, $Rv_{r,t}$ is sorted in descending order by r_t , and $i_{j,r}$ is sorted in descending order by $Ri_{i,r}^t$. In the decoding process, the AGVs are first assigned to the routes in the order of $Rv_{r,t}$. Then the type and quantity of parts to be distributed are determined according to the order of $Ri_{i,r}^t$. Finally, the workstation inventories in production cycle $t + 1$ are determined to ensure that there will be no inventory shortage.

4.3. Diagonally linear random initialization strategy

For meta-heuristic algorithms, the selection of the initial population can effectively affect the rate of convergence (Rauf et al., 2020). Adopting an appropriate population initialization method can not only obtain a large number of initial populations in a short time but also obtain high-quality initial populations, so as to improve the performance of the algorithm. In order to quickly obtain the initial population with good diversity and uniformity, this paper introduced the diagonal linear initialization strategy to generate the initial population according to the problem attribute (Li et al., 2021). The pseudocode for the diagonally linear random initialization strategy is shown in Fig. 5. This strategy is not only easy to operate but can also greatly improve the diversity and uniformity of the initial population. More importantly, its performance does not decline as the number of dimensions increases, so it can perform well in multi-objective problems.

4.4. Search strategy based on AEFA

AEFA is a population-based meta-heuristic algorithm proposed by Anita and Yadav (2019). Inspired by Coulomb's law, AEFA simulates the motion of charged particles in an electrostatic field. Therefore, the search strategy based on AEFA is described as follows. In the search space, each charged particle represents a feasible solution, whose strength is measured by the amount of charge it carries. The larger the amount of charge, the closer the particle is to the theoretical optimal value. Each charged particle is acted upon by the attraction of other particles, which, according to the laws of motion, can form a net force and an acceleration in that direction. The velocity and position of the charge can be calculated from the acceleration of the particle. The positions and fitness functions of charged particles determine their charge and unit mass. Finally, when the amount of a certain charge in the search space is large, other charges will be attracted by the large charge so as to get closer to the direction of the large charge. Thus, the algorithm converges to the optimal solution. Therefore, the process of searching for the optimal solution can be visualized as the process of the charged particle with a large charge attracting all the other particles with smaller charges to it.

The specific steps of the algorithm are shown as follows.

Step 1 In the search space, the population of candidate solutions (charged particles) is initialized, including the position X , velocity V and charge Q of charged particles.

In the d -dimensional space, the position X_i of particle i is expressed by equation (21).

$$X_i = (x_i^1, x_i^2, \dots, x_i^d), \quad \forall i = 1, 2, \dots, N \quad (21)$$

Where, x_i^d represents the position of particle i in the d -dimensional space, and N represents the total number of particles.

The initial velocity V of each charged particle is 0. The charge of each particle is Q_i , and the initial charge of each particle is equal.

$$Q_i(t = 0) = Q, \quad \forall i = 1, 2, \dots, N \quad (22)$$

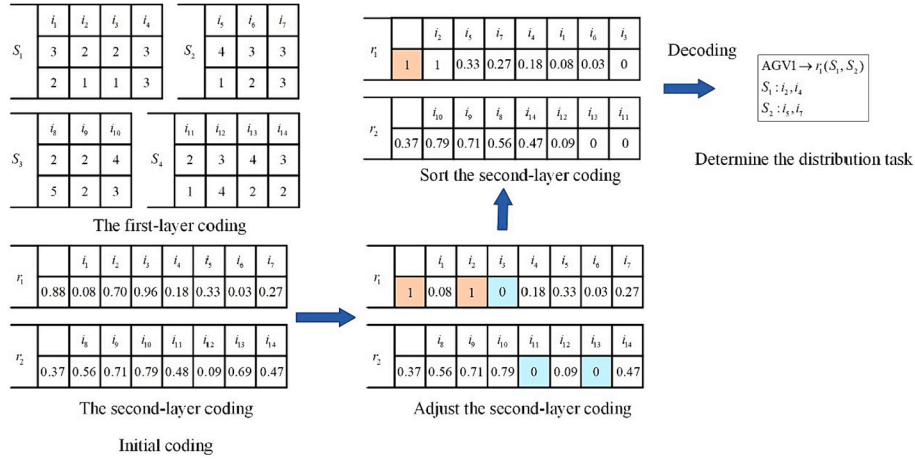


Fig. 4. Schematic diagram of decoding process.

Table 2

Parameters of example.

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}	i_{13}	i_{14}
$lvnk_{j,s}^t$	2	1	2	2	2	2	2	1	1	2	2	2	4	2
$lv_{j,s}^t$	3	1	2	6	2	3	6	5	2	4	2	8	7	4
$nk_{j,s}$	3	2	2	3	4	3	3	2	2	4	2	3	4	3
$cb_{j,s}$	2	1	1	3	1	2	3	5	2	3	1	4	2	2
d_j^t	1	1	0	0	1	1	0	1	0	1	0	1	0	0
d_j^{t+1}	0	1	0	1	0	1	1	1	0	1	1	0	1	0

Pseudocode: Diagonal Linear Random Initialization

- 1: **Input:** the range $[s_d^{min}, s_d^{max}]$ for each dimension s_d , the population size N , the dimension of problem D .
- 2: $S_{init} = \text{zeros}(D, N)$ // allocate memory space
- 3: **for** $d = 1, 2, \dots, D$ **do**
- 4: **for** $j = 2, \dots, J - 1$ **do**
- 5: $ds = \frac{s_d^{max} - s_d^{min}}{N}$ // length of each interval in d -th dimension
- 6: $S_{init}(d, 1) = s_d^{min} + rand \times ds$,
- 7: $S_{init}(d, j) = s_d^{min} + (j - 1) \times ds + rand \times ds$
- 8: $S_{init}(d, J) = s_d^{max}$
- 9: **end for**
- 10: **end for**
- 11: **return** S_{init} as a vector

Fig. 5. Pseudocode of diagonally linear random initialization strategy.

Where, Q is a set constant.

AEFA is a memory-based algorithm. So, it uses the global optimum and individual optimum fitness of each particle over all iterations to update its position. In iteration $t + 1$, the optimal fitness value $P_i^d(t + 1)$ obtained by particle i in the d -dimensional space is expressed by equation (23).

$$P_i^d(t + 1) = \begin{cases} P_i^d(t), & \text{if } f(P_i(t)) < f(X_i(t + 1)) \\ x_i^d(t + 1), & \text{else} \end{cases} \quad (23)$$

Where, P_i represents the optimal fitness value of particle i .

Step 2 Calculate the fitness value fit for each particle.

The traditional AEFA fitness evaluation method is not suitable for multi-objective problems, while Pareto domination is a common way to judge the merits and disadvantages of low-dimensional multi-objective optimization individuals. Therefore, a fitness evaluation method based on Pareto domination is introduced to evaluate the fitness of the algorithm, so as to guide the search process. The domination-based fitness evaluation method is represented by equation (24).

$$fit_i = DomN_i + \frac{1}{N} + 1 \quad (24)$$

Where, $DomN_i$ represents the number of other particles in the population dominated by particle i . The optimal fitness value fit_{best} and the worst fitness value fit_{worst} can be calculated by Eqs. (25) and (26).

$$fit_{best}(t) = \max(fit_i(t)), i \in (1, 2, \dots, N) \quad (25)$$

$$fit_{best}(t) = \min(fit_j(t)), j \in (1, 2, \dots, N) \quad (26)$$

Because this paper needs to calculate the minimum value of the objectives, fit_{best} is the maximum fit obtained from particle i .

Step 3 Calculate the coulomb constant K of the particles.

K is determined by the current iteration and the maximum number of iterations. In order to control the search accuracy of the algorithm, it decreases exponentially. In iteration t , $K(t)$ can be calculated by Eq. (27).

$$K(t) = K_0 * \exp(-\tau^{ite/ite_{max}}) \quad (27)$$

Where, K_0 represents the initial value, τ is a constant, ite represents the current iterations, and ite_{max} represents the maximum number of iterations.

At the initial stage of the algorithm, K is a higher value to facilitate the initial exploration of the algorithm. Then, the search precision of the algorithm is controlled by decrementing K .

Step 4 Calculate the charge Q of the particles.

In iteration t , the charge $Q_i(t)$ of particle i can be calculated through the fitness value, which is expressed by Eqs. (28) and (29).

$$Q_i(t) = q_i(t) / \sum_{i=1}^N q_i(t) \quad (28)$$

$$q_i(t) = \exp(fit_i(t) - fit_{worst}(t) / fit_{best}(t) - fit_{worst}(t)) \quad (29)$$

Where $q_i(t)$ helps determine the total amount of charge interacting with charge i in iteration t .

Step 5 Calculate the coulomb force F of the particles.

In iteration t , particle j is subject to the coulomb force $F_{ij}^d(t)$ of particle i in d -dimensional space, which is expressed by Eq. (30).

$$F_{ij}^d(t) = K(t)Q_i(t)Q_j(t)(P_j^d(t) - X_i^d(t)) / R_{ij}(t) + \xi \quad (30)$$

Where, ξ is a minuscule constant, R_{ij} means the Euclidean distance between particle i and particle j , which can be calculated by Eq. (32). Therefore, the total force $F_i^d(t)$ acting on particle i in iteration t is expressed by Eq. (31).

$$F_i^d(t) = \sum_{j=1, j \neq i}^N rand_j F_{ij}^d(t) \quad (31)$$

Where, $rand_j$ represents the random number between $[0, 1]$. In iteration t , the Euclidean distance $R_{ij}(t)$ of particle i and particle j is expressed by Eq. (32).

$$R_{ij}(t) = \|X_i(t), X_j(t)\|_2 \quad (32)$$

Step 6 Calculate the acceleration a .

The acceleration $a_i^d(t)$ of particle i in d -dimensional space in iteration t is expressed by equation (33).

$$a_i^d(t) = F_i^d(t) / Q_i(t) \quad (33)$$

Step 7 Update particle velocity V and position X .

In iteration $t + 1$, the velocity $V_i^d(t + 1)$ and position $X_i^d(t + 1)$ of particle i in d -dimensional space is updated according to the acceleration a_i^d . The update modes are expressed by equations (34) and (35).

$$V_i^d(t + 1) = rand_i * V_i^d(t) + a_i^d(t) \quad (34)$$

$$X_i^d(t + 1) = X_i^d(t) + V_i^d(t + 1) \quad (35)$$

Where, $V_i^d(t)$ represents the velocity of particle i in iteration t , $X_i^d(t)$ represents the position of particle i in iteration t , and $rand_i$ represents the random number between $[0, 1]$.

Step 8 Determine whether the stopping condition is satisfied. If so,

output the optimal value, otherwise repeat steps 2–8.

4.5. Local optimization strategy based on the SARSA selection mechanism

In the basic AEFA, only K in equation (27) is used to conduct local search, which leads to the local optimization of the algorithm easily. Therefore, to improve the local search ability of the algorithm, four optimization operators are designed to optimize the algorithm. Local optimization repeats NLS times, and the SARSA mechanism is used to select the best local optimization scheme, so as to effectively improve the local search ability of the algorithm. Among them, the SARSA is a reinforcement learning algorithm proposed by Rummery in 1994 (Rummery & Niranjan, 1994). The SARSA uses the actual Q value to update, which is carried out in the order of $(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1})$. The specific update mode is represented by equation (36).

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)) \quad (36)$$

Where, s_t represents the current state, a_t represents the action selected under the current state, r_{t+1} represents the reward obtained after selecting action a_t , s_{t+1} and a_{t+1} respectively represent the subsequent action, $Q(s_t, a_t)$ represents the expected reward value of the action a_t selected by state s_t , $\alpha \in [0, 1]$ is the learning rate, γ is the attenuation factor, which represents the decay of future reward. The pseudocode of the SARSA selection mechanism is shown in Fig. 6.

4.5.1. Calculation method of states

In reinforcement learning, state s represents the environmental condition of the action. For the proposed problem, s describes the performance of the population based on action a . In this paper, the performance index hypervolume (HV) is used as the evaluation method of s . HV is a comprehensive index used to evaluate the distribution pattern and convergence of non-dominated solution sets in multi-objective optimization. The calculation method is shown in equation (37).

$$HV(s) = \delta(\sum_{ns \in |NS|} v_{ns}) \quad (37)$$

Where, δ represents the Lebesgue measure, v_{ns} represents the HV containing the non-dominated solution ns and the reference point, $|NS|$ represents the number of non-dominated solutions. The reference point in this paper is $(1, 1)$. Then, the larger the value of HV, the better the Pareto frontier.

4.5.2. Calculation method of actions

To improve the local search ability of the algorithm, four local search operators are designed, $LS = \{OBL, RCO, LIO, ECO\}$.

(1) Opposition-based learning optimization operator (OBL)

As shown in Fig. 7(a), for the current individual $X_i = \{X_{i,1}, X_{i,2}, \dots, X_{i,D}\}$, the concept of the reverse point is defined. $OX_i = \{OX_{i,1}, OX_{i,2}, \dots, OX_{i,D}\}$ are the reverse points of current solution X_i . The elements in the reverse point can be represented by equation (38).

$$OX_{i,d} = X_{i,d}^{lb} + X_{i,d}^{ub} - X_{i,d} \quad (38)$$

Where, $X_{i,d}^{lb}$ represents the lower bound of particle i in d -dimensional space, and $X_{i,d}^{ub}$ represents the upper bound of particle i in d -dimensional space.

(2) Random cross optimization operator (RCO)

As shown in Fig. 7(b), the current individual X_i^t and the optimal individual X_{best}^t of the population are taken as parent 1 and parent 2. Two columns in the code are arbitrarily selected as the crossover boundary, and the coding columns outside the crossover boundary of parent 1 and

Pseudocode: SARSA mechanism

```

1: Input  $Q(s,a)$ ,  $\alpha$ ,  $\gamma$ 
2: For  $ite = 1, 2, \dots, ite_{max}$  do
3:   Initialize  $s$ 
4:   Choose  $a$  from  $s$  using policy derived from  $Q$  according to  $\epsilon$ -greedy
5:   While  $s$  is terminal do
6:     Take action  $a$ , observe  $r$  and  $s'$ 
7:     Choose  $a'$  from  $s'$  using policy derived from  $Q$  according to  $\epsilon$ -greedy
8:      $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s,a') - Q(s,a)]$ 
9:      $s \leftarrow s'$ ,  $a \leftarrow a'$ 
10:  end while
11: end for
12: return  $s$ ,  $a$ 

```

Fig. 6. Pseudocode of the SARSA selection mechanism.

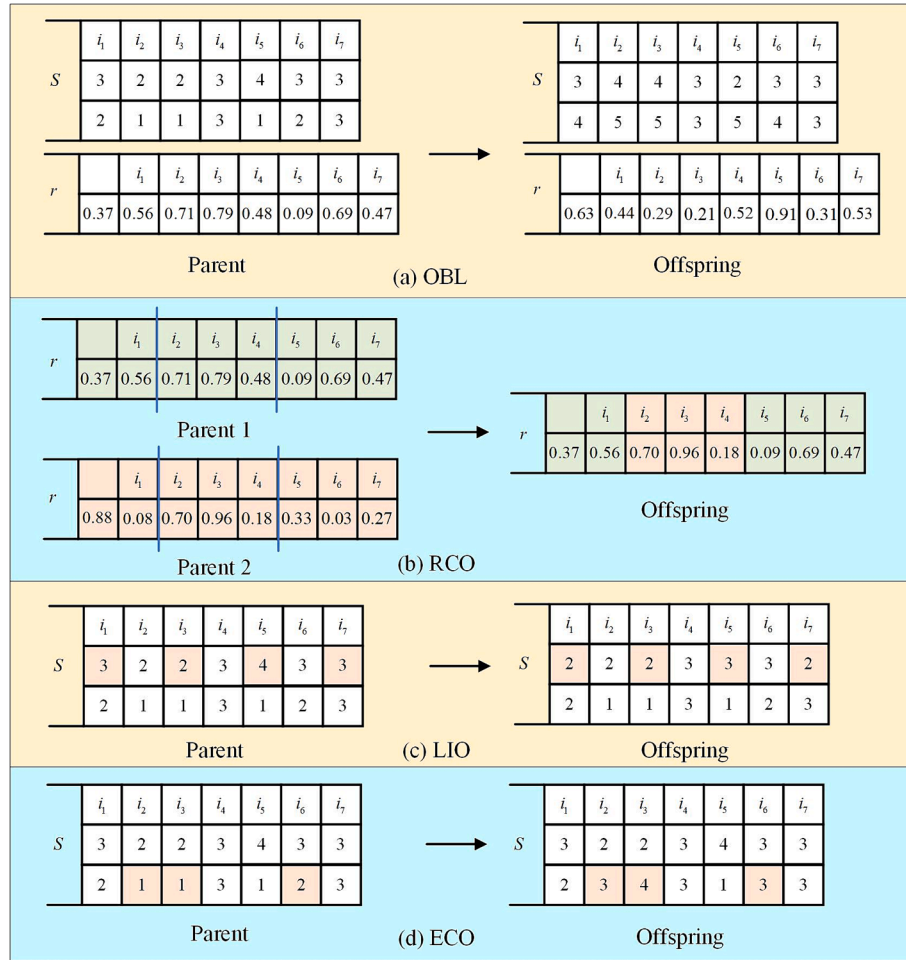


Fig. 7. Local search operators.

the coding columns inside the crossover boundary of parent 2 are reserved to offspring individual X_i^{t+1} .

(3) Line-side inventory optimization operator (LIO)

As shown in Fig. 7(c), the Kanban quantity for each part needs to be minimized in order to keep the line-side inventory as small as possible. Therefore, $nk_{j,s}$ in the first-layer coding of individual X_i^t can be adjusted.

Randomly select N parts to adjust the coding according to equation (39), so as to obtain a new offspring individual X_i^{t+1} .

$$nk_{j,s}^{t+1} = \left[rand \cdot (nk_{j,s}^t - nk_{j,s}^{min}) + nk_{j,s}^{min} \right] \quad (39)$$

Where, $\lceil \cdot \rceil$ represents rounded down and $rand$ represents the random number between $[0, 1]$.

(4) Energy consumption optimization operator (ECO)

As shown in Fig. 7(d), to minimize the energy consumption of AGVs, the number of distributions needs to be reduced as much as possible, so the material bin capacity of parts can be increased. Therefore, $cb_{j,s}$ in the first-layer coding of individual X_i^t can be adjusted. Randomly select N parts to adjust the coding according to equation (40), so as to obtain new offspring individual X_i^{t+1} .

$$cb_{j,s}^{t+1} = \left\lfloor \text{rand} \bullet (cb_{j,s}^{\max} - cb_{j,s}^t) + cb_{j,s}^t \right\rfloor \quad (40)$$

Where, $\lceil \cdot \rceil$ represents rounded upward and rand represents the random number between (0, 1).

4.5.3. Calculation method of γ

Immediate reward γ represents a reward or punishment for performing action a . Action a should be rewarded when the offspring individual is better, and action a should be punished when the parent individual is better. Therefore, the calculation method of γ is shown in equation (41).

$$\gamma = f_{Rr}(Rr) \bullet f_{HV}(HV) \quad (41)$$

Where, $f_{Rr}(Rr)$ represents the evolution degree of the offspring individual, which is expressed by equation (42). $f_{HV}(HV)$ is the amount of reward or punishment corresponding to the population performance, which is expressed by equation (44).

$$f_{Rr}(Rr) = \begin{cases} -1, & Rr < 0 \\ 0, & Rr = 0 \\ 1, & Rr > 0 \end{cases} \quad (42)$$

Where, Rr represents optimization rate, which can be calculated by equation (43).

$$Rr = \frac{\sum_{i \in NP_1} \prod_{j \in m} f_j(x_i) / |NP_1| - \sum_{i \in NP_2} \prod_{j \in m} f_j(y_i) / |NP_2|}{\sum_{i \in NP_1} \prod_{j \in m} f_j(x_i) / |NP_1|} \quad (43)$$

Where, NP represents the population size, m represents the number of objectives, x_i and y_i represent the feasible solutions in the current population and the previous generation population, respectively.

$$f_{HV}(HV) = \begin{cases} 0.5, & HV \in [0, 0.3] \text{ and } Rr < 0, \text{ or } HV \in [0.7, +\infty) \text{ and } Rr > 0 \\ 1, & HV \in [0.3, 0.7] \\ 1.5, & HV \in [0.7, +\infty) \text{ and } Rr < 0, \text{ or } HV \in [0, 0.3] \text{ and } Rr > 0 \end{cases} \quad (44)$$

Therefore, there are 6 possible values of γ , including $\gamma = \{-1.5, -1, -0.5, 0, 0.5, 1, 1.5\}$.

4.5.4. Calculation method of adaptive learning rate α

Learning rate α , represents the proportion of newly learned information, is used to control the proportion of future rewards in the learning process. A higher learning rate indicates that the algorithm tends to make use of newly learned information, while a lower learning rate indicates that the algorithm tends to make use of existing information. The adaptive learning rate is adopted in this paper. At the early stage of algorithm iteration, a higher learning rate is conducive to accelerating the algorithm convergence speed. With the progress of iteration, the learning rate should be gradually reduced to improve the exploitation performance. The update of α is shown in equation (45).

$$\alpha = 1 - (0.9 \times \text{ite} / \text{ite}_{\max}) \quad (45)$$

Where, ite represents the current iteration and $\text{ite}_{\max}$ represents the maximum number of iterations.

4.6. Selection strategy based on Pareto domination

To keep the population size unchanged after each iteration, it is necessary to select the pre-NP individuals as the new parent population from the mixed population of parent and offspring. Then, the Pareto level and crowding distance of NSGA-II are used to select the offspring population. The general principle of selection is as follows. Individuals with low Pareto levels are preferentially selected. If multiple individuals have the same Pareto level, individuals with large crowding distances are preferred.

4.6.1. Pareto level sorting

Firstly, the set d_i and number n_i of individuals dominated by each individual i in the population are calculated. Individuals with $n_i = 0$ are placed into the set $Level1$ and remove their dominant relationships over other individuals. Then, update the n values of all remaining particles according to the newly obtained d_i . Similarly, the individuals with $n_i = 0$ now should be put into the set $Level2$, repeat the above steps until all the individuals in the population have been graded.

4.6.2. Crowding distance

The crowding distance is a density index that represents the density of each individual at the same Pareto level. The diversity of solution space is maintained by preserving solutions with less similarity. The crowding distance CD_i of individual i is calculated as follows.

$$CD_i = \sum_{j \in M} p_j(i+1) - p_j(i-1) / p_j^{\max} - p_j^{\min} \quad (46)$$

Where, M represents the number of objectives, $p_j(i+1)$ and $p_j(i-1)$ respectively represent the objective values of two adjacent individuals of individual i in objective j , $p_j^{\max}$ and $p_j^{\min}$ respectively represent the maximum and minimum values of all individual objective j values.

4.7. Repair mechanism

The proposed algorithm will generate infeasible solutions during the iterative process, which can be divided into two situations. In the first situation, the reason for the infeasible solution is that the coding value will exceed the coding range ($cod_{\min}$, $cod_{\max}$) during the iteration. Therefore, repair mechanism 1 is defined to regenerate the over-range coding value within the coding range. The second situation is that the coding value does not satisfy the constraints (19). For problems that do not satisfy constraint (19), repair mechanism 2 is defined. If part i belonging to workstation s does not meet the constraint (19), $nk_{j,s}^i = \left\lceil \text{rand}^* (nk_{j,s} - nk_{j,s}^{\min}) + nk_{j,s}^{\min} \right\rceil$ is used to regenerate $nk_{j,s}$ value for part i . Then, determine whether the constraint is satisfied again, if not, repeat

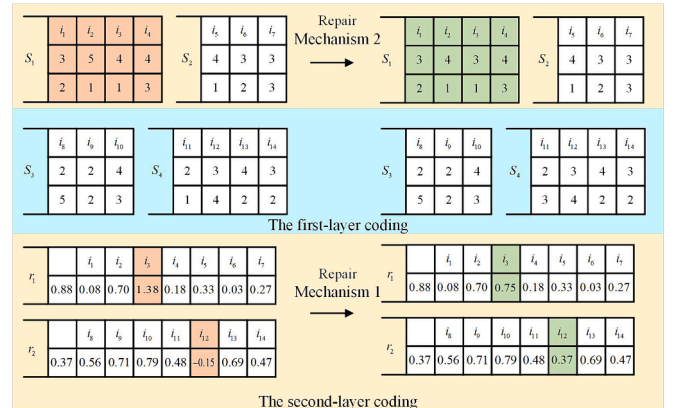


Fig. 8. Principle of algorithm repair mechanisms.

the above operations.

Fig. 8 shows the principle of the two repair mechanisms. For example, the second-layer coding exceeds the upper and lower bounds of coding, repair mechanism 1 can be used to repair this. In addition, in the first-layer coding, the coding value of workstation S_1 did not meet constraint (19), which can be repaired by repair mechanism 2.

5. Computational experiments

In this paper, a MOAEFASA algorithm is designed to solve the MMDSPCE. To evaluate the performance of the algorithm, computational experiments are carried out. The proposed algorithm is compared with the Epsilon constraint methods, NSGA-II, improving the strength Pareto evolutionary algorithm (SPEA-II), multi-objective equilibrium optimization algorithm (MOEO), and multi-objective neural network algorithm (MONNA) in small-scale problems. Then, it is compared with NSGA-II, SPEA-II, MOEO and MONNA in medium-scale and large-scale problems. In addition, the optimization strategies of the algorithm are tested. All computational experiments are implemented on Gurobi and Matlab 2021a. The PC is configured with an AMD R7-5800H CPU, 16 GB RAM, and 3.2 GHz.

5.1. Experimental settings

In a group of instances, we refer to the benchmark example of the traditional automobile assembly line material distribution problem given by Fathi et al. (2015). The relevant parameters of this experiment are described as follows. MMALs assemble three models simultaneously, each with a different bill of materials. In production cycle t , the material demand d_{st} of station s is the uniformly distributed random number in the interval $[1, 3]$, where $\lceil \bullet \rceil$ is rounded to the nearest integer. The Kanban quantity and the material bin capacity for each part are less than 6. The weight of each part is a random number evenly distributed in the interval $[1, 5]$. The distance between adjacent stations is 8m. Besides, $T = 90s$, $w = 2m/s$, $Cv = 6$, $we_v = 5$, $tl = 10s$, $tr = 5s$.

In this paper, three scenarios of small-scale problems, medium-scale problems and large-scale problems are designed. Four algorithms are used to compare with the proposed MOAEFASA in all the three scale problems. In small-scale problems, $T = \{8, 10, 15\}$, $|S| = \{4, 6, 8, 10\}$. The rest of the relevant parameters are set as follow. Route number $|R| = 2$. Therefore, the number of workstations passed by each route $RS \in \{2, 3, 4, 5\}$, correspondingly. $|V| = 2$. The initial inventory of each part is 4. For small-scale problems, the results obtained by the five algorithms are compared with those obtained by the Epsilon constraint method. However, the Epsilon constraint method is not suitable for solving medium-scale and large-scale problems because Gurobi will take significantly longer time to solve problems as the problem scale increases. In medium-scale problems, $T = \{30, 40, 50\}$, $|S| = \{30, 40, 50\}$. In large-scale problems, $T = \{60, 80, 100\}$, $|S| = \{60, 80, 100\}$. The rest of the relevant parameters are setting as follow. $RS = 5$. Then, $|R| = |S|/RS$. The initial inventory of each part is full. The number of AGVs is calculated by theorem 1. For medium-scale and large-scale problems, five algorithms are used to solve and compare.

5.2. Performance measures

In order to evaluate the performance of multi-objective algorithms, it is necessary to conduct a comprehensive evaluation of the Pareto sets obtained by each algorithm. In this paper, the following three performance metrics are selected to evaluate the Pareto optimality, uniformity, distribution and convergence of the solution. Given that HV has been employed as an evaluation metric in the local search strategy, it is no longer utilized for algorithm evaluation.

- (1) The degree of Pareto optimality of the solution set (DPO)

$$DPO_l = |NP_l^*|/NP_l \quad (47)$$

Where, NP_l^* represents the number of solutions that are non-dominated by the Pareto solution sets of the other two algorithms among the NP_l Pareto solutions obtained by algorithm l . Therefore, the larger the value of DPO_l , the better the quality of the Pareto solution obtained by algorithm l , which means the performance of algorithm l is better.

- (2) The evenness of the distribution (ES).

$$ES_l = 1/\sqrt{D} \sqrt{1/|NP_l| \sum_{x_i \in NP_l} (D_i - \bar{D})^2} \quad (48)$$

Where, $\bar{D} = \frac{1}{|NP_l|} \sum_{x_i \in NP_l} D_i$, D_i represents the Euclidean distance between x_i and its nearest neighbor solution. Therefore, the smaller the value of ES_l , the more evenly distributed the solution set obtained by algorithm l , which means the performance of algorithm l is better.

- (3) The inverted generational distance (IGD).

$$IGD_l = \sum_{\omega \in l} d(\omega, l^*)/|l^*| \quad (49)$$

Where, ω represents a solution in the Pareto solution set obtained by algorithm l ; $d(\omega, l^*)$ is the minimum Euclidean distance between the solution ω and the solution l^* in the actual Pareto solution set, $|l^*|$ represents the number of individuals on the actual Pareto front. Therefore, the smaller the value of IGD_l , the smaller the sum of the average distances between the Pareto solution set obtained by algorithm l and the actual Pareto solution set, which means the convergence and distribution of algorithm l are better.

5.3. Parameters settings

The solution performance of MOAEFASA is greatly affected by the parameters. Therefore, Taguchi's experiments are used to optimize the algorithm parameters. Parameters that need to be tuned in MOAEFASA are the initial coulomb constant k_0 , the control factor of search accuracy τ and the number of local searches NLS . Each parameter contains three levels. $k_0 \in \{100, 150, 200\}$, $\tau \in \{40, 60, 80\}$, $NLS \in \{1, 5, 10\}$. Taguchi $L9(3^3)$ orthogonal array is used to find the optimal parameter combination for the experiments.

According to the $L9$ orthogonal table, 9 groups of experiments are generated for each problem scale. Each experiment is repeated 10 times. To compare the optimization performance of the algorithm under different parameter levels, DPO is used as the evaluation metric, which represents the proportion of the current solution that is not dominated by other solutions obtained from the other eight groups of experiments. Obviously, the larger the value of DPO, the higher the quality of the solutions. Then, the corresponding parameters are the optimal parameters.

Table 3 shows the experimental results generated by Minitab. It can be seen from the table that different problem scales have different optimal parameter combinations. In small-scale problems, the optimal parameter combination is $k_0 = 100$, $\tau = 40$, $NLS = 5$. In medium-scale problems, the optimal parameter combination is $k_0 = 150$, $\tau = 40$, $NLS = 5$. In large-scale problems, the optimal parameter combination is $k_0 = 150$, $\tau = 40$, $NLS = 10$.

In addition, the proposed MOAEFASA is compared with four algorithms with excellent computational performance. The first is MOEO, a recent multi-objective intelligent search algorithm that has been shown to perform well in solving a variety of problems (Premkumar et al., 2022). The second is MONNA, which has been designed to obtain recommendations for optimal electric vehicle charging station infrastructure based on photovoltaic systems (Mehouachi et al., 2023). The other

Table 3
Analyses results of the Taguchi's experiments.

Problem scale	Level	Parameters		
		k_0	τ	NLS
Small	1	0.185	0.173	0.086
	2	0.123	0.074	0.185
	3	0.025	0.086	0.062
	Del	0.160	0.099	0.123
	Rank of major factors	1	3	2
	Optimal level	1	1	2
	Optimal combination	100	40	5
Medium	1	0.097	0.161	0.108
	2	0.172	0.075	0.140
	3	0.065	0.097	0.086
	Del	0.108	0.086	0.054
	Rank of major factors	1	2	3
	Optimal level	2	1	2
	Optimal combination	150	40	5
Large	1	0.067	0.143	0.067
	2	0.171	0.086	0.105
	3	0.095	0.105	0.162
	Del	0.105	0.057	0.095
	Rank of major factors	1	3	2
	Optimal level	2	1	3
	Optimal combination	150	40	10

two are the classical multi-objective evolutionary algorithms NSGA-II and SPEA-II, which are widely used to solve a variety of complex multi-objective problems and are often used as control objectives to validate the performance of the algorithms. Therefore, DOE experiments are carried out for the four comparison algorithms. Then, the optimal parameters of the four algorithms under three scenarios are obtained.

Table 4
Parameter settings of the metaheuristic algorithms.

Algorithms	Parameters	Parameter levels		
		Low	Medium	High
MOAEFASA	Number of populations	100	100	100
	Initial Coulomb constant	100	150	150
	Search accuracy control factor	40	40	40
	Local search	5	5	10
	Maximum generation	100	100	100
MOEO	Number of populations	100	100	100
	Global search weight factor	1.5	1.5	2
	Iterative control factor	1.5	1.5	1
	Generation rate control factor	0.5	0.4	0.4
	Maximum generation	100	100	100
NSGA-II	Number of populations	100	100	100
	Crossover probability	0.7	0.8	0.8
	Mutation probability	0.15	0.2	0.2
	Maximum generation	100	100	100
MONNA	Number of populations	100	100	100
	Archive size	10	50	50
	Grid inflation	0.1	0.1	0.1
	Number of grids per each dimension	10	10	20
	Leader selection pressure	4	4	4
	Extra repository member selection pressure	2	2	2
	Maximum generation	100	100	100
SPEA-II	Number of populations	100	100	100
	Archive size	10	50	50
	Crossover probability	0.6	0.6	0.7
	Mutation probability	0.15	0.25	0.3
	Maximum generation	100	100	100

The final parameter settings of the five algorithms are shown in Table 4.

5.4. Small-scale comparison experiments

Firstly, Gurobi is used to solve small-scale problems based on the Epsilon constraint method (Yang et al., 2019). The Epsilon constraint method is the most commonly used exact algorithm for solving bi-objective problems. The principle of the method is to convert one of the objectives into a constraint, so that the problem becomes a single-objective problem with Epsilon constraints. Then by gradually reducing the value of Epsilon, the relationship between the two objectives is established. Finally, the optimal Pareto frontier is obtained.

In this paper, the parameters of the small-scale problems are $T = \{8, 10, 15\}$, $|S| = \{4, 6, 8, 10\}$, which can be combined to generate 12 instances in total. The five algorithms are independently run 10 times for each instance. In order to obtain a uniform and effective solution in a short time, the scale SL of Pareto frontier solution set is set as 10. The upper bound of the running time of Gurobi is 3600 s. The results of Pareto frontier comparison and CPU runtime comparison are given by Fig. 9 and Table 5.

Fig. 9 shows the Pareto frontier obtained by six methods. It can be clearly seen that for small-scale problems, the five meta-heuristic algorithms can obtain high-quality Pareto solutions. In most instances, the Pareto front obtained by MOAEFASA is closer to that obtained by the Epsilon-constraint method, indicating that the quality of the Pareto solution set obtained by MOAEFASA is better. Then, although the performance of decreases with the increase in problem scale, the deviation of the results is acceptable.

Table 5 shows the results for CPU runtime. The results show that the running time of Gurobi increases significantly as T and S increase. When $T = 10, S = 10$ and $T = 15, S = 10$, Gurobi cannot solve the problem in the specified time. On the contrary, the five meta-heuristic algorithms can obtain the solution of the problem in a short time. Therefore, the meta-heuristic algorithms are better than the Epsilon-constraint method in terms of computational efficiency.

5.5. Comparison of experimental results of different algorithms

To further evaluate and verify the effectiveness of the proposed algorithm MOAEFASA, it is compared with the other four meta-heuristic. First, the Pareto frontiers of the five algorithms are compared in 9 instances under medium-scale and large-scale problems, respectively. Each experiment is repeated 10 times. Pareto frontier integrated the results obtained from 10 experiments, which are shown in Fig. 10. As can be seen from Fig. 10, in both medium-scale and large-scale problems, MOAEFASA can obtain better solutions than the comparison algorithms, and the solutions obtained by the comparison algorithms are mostly dominated by those obtained by MOAEFASA. Fig. 11 shows in more detail the Pareto frontier for each algorithm in instance 1 of Fig. 10.

Then, the MOAEFASA is systematically evaluated in 18 instances. Based on the scale of S and T , the 18 instances are divided into two groups, corresponding to medium-scale and large-scale problems respectively. Each group contained 9 instances. For each instance, the five algorithms run 10 times independently. The Pareto frontier combines the results of 10 experiments. The DPO, ES and IGD values obtained are averaged and used as the final evaluation metrics of the algorithms. Table 6–8 show the experimental results of the five algorithms under different problem scales, and the optimal value of each instance is marked in bold. Based on the experimental results, the following conclusions can be obtained.

- (1) First, the DPO values of the five algorithms are compared. On average, the calculation results of the five algorithms have obvious differences. The DPO values obtained by MOAEFASA are significantly better than the four compared algorithms. In

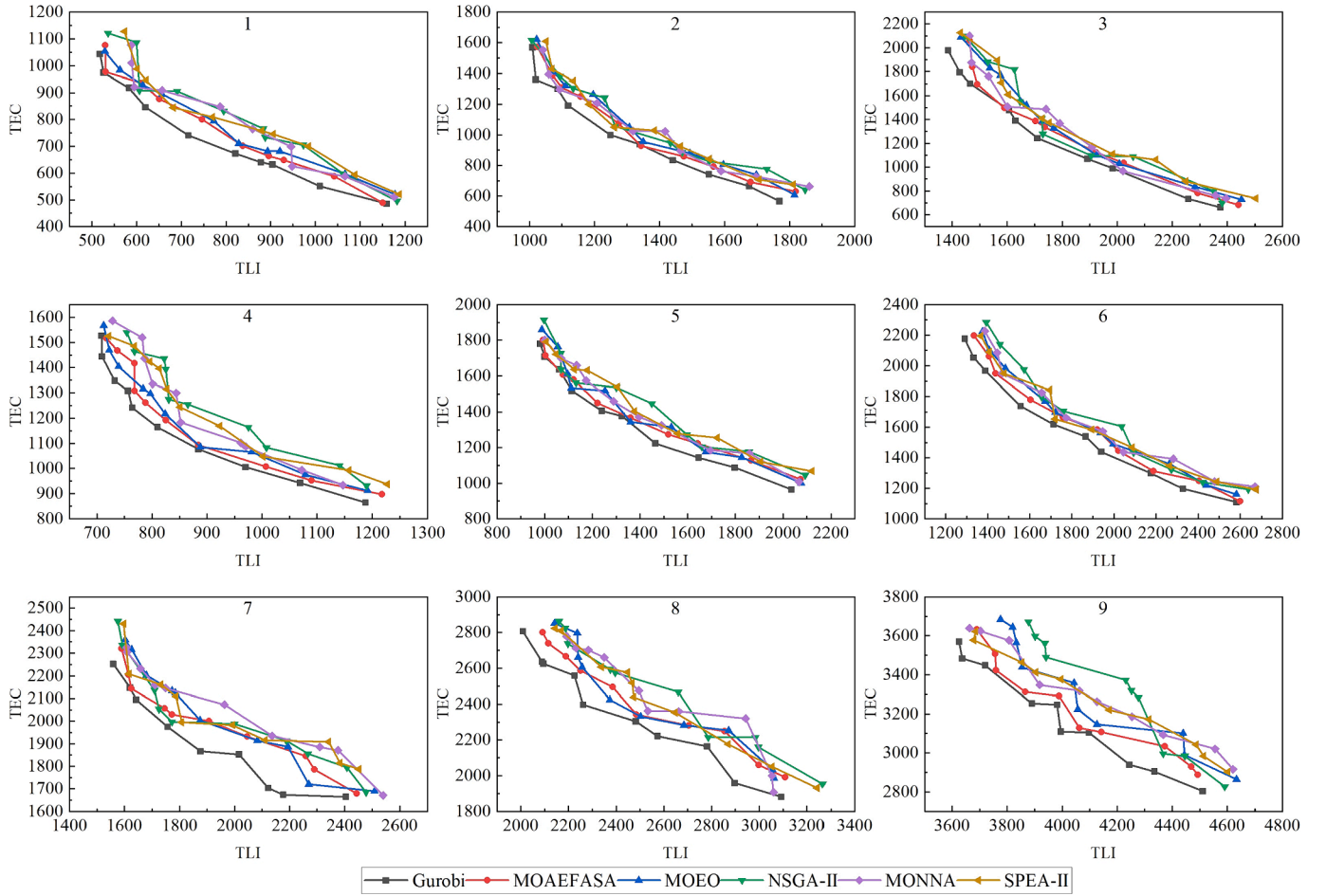


Fig. 9. Pareto frontiers obtained by Gurobi and meta-heuristic algorithms.

Table 5

The CPU time of Gurobi and meta-heuristic algorithms.

No.	T	S	Gurobi	MOAEFASA	MOEO	NSGA-II	MONNA	SPEA-II
1	8	4	27.3	2.4	1.9	1.2	1.9	1.3
2	8	6	64.9	5.4	3.8	2.0	3.4	2.2
3	8	8	134.3	6.2	4.7	2.6	4.7	2.6
4	8	10	328.3	6.9	5.4	3.3	4.9	3.3
5	10	4	418.9	3.7	3.2	2.1	2.9	2.4
6	10	6	1090.4	5.2	4.6	2.8	4.5	2.9
7	10	8	2144.9	8.1	5.6	3.4	5.5	3.4
8	10	10	–	9.8	6.2	4.3	6.0	4.6
9	15	4	798	7.5	5.5	3.5	5.0	3.7
10	15	6	2115.2	8.4	4.9	4.2	4.5	4.3
11	15	8	3570.5	8.9	6.7	6.5	6.5	6.4
12	15	10	–	10.5	9.3	7.4	8.4	7.8

addition, among the 18 instances, the optimal DPO ratio obtained by MOAEFASA is 14/18. The optimal DPO ratio obtained by MOEO and MONNA, SPEA-II and NSGA-II are 2/18, 2/18, 0 and 0, correspondingly. Therefore, the MOAEFASA has a significant advantage in the number of Pareto solutions obtained.

- (2) Then, the ES values of the five algorithms are compared. On average, the ES value obtained by MOAEFASA is the smallest. The ES values obtained by MONNA and SPEA-II have small deviations and are closer to that obtained by MOAEFASA. Then the ES value obtained by MOEO is the largest. In addition, among the 18 instances, the ratio of optimal ES values obtained by MOAEFASA is 10/18. The ratio of optimal ES values obtained by SPEA-II, MONNA, MOEO and NSGA-II are 6/18, 2/18, 0 and 0,

correspondingly. The results show that MOAEFASA is more likely to obtain a uniformly distributed Pareto set than the comparison algorithms.

- (3) Finally, the IGD values of the five algorithms are compared. On average, MOAEFASA has the lowest average IGD values which is significantly lower than those obtained by all the comparison algorithms. The lowest IGD value in the comparison algorithms is obtained for MOEO, which is about 48.9 % higher than that of MOAEFASA. In addition, among the 18 instances, the ratio of optimal IGD values obtained by MOAEFASA is 13/18. The ratio of optimal IGD values obtained by MOEO, MONNA, SPEA-II and NSGA-II are 3/18, 2/18, 0 and 0, correspondingly. The results

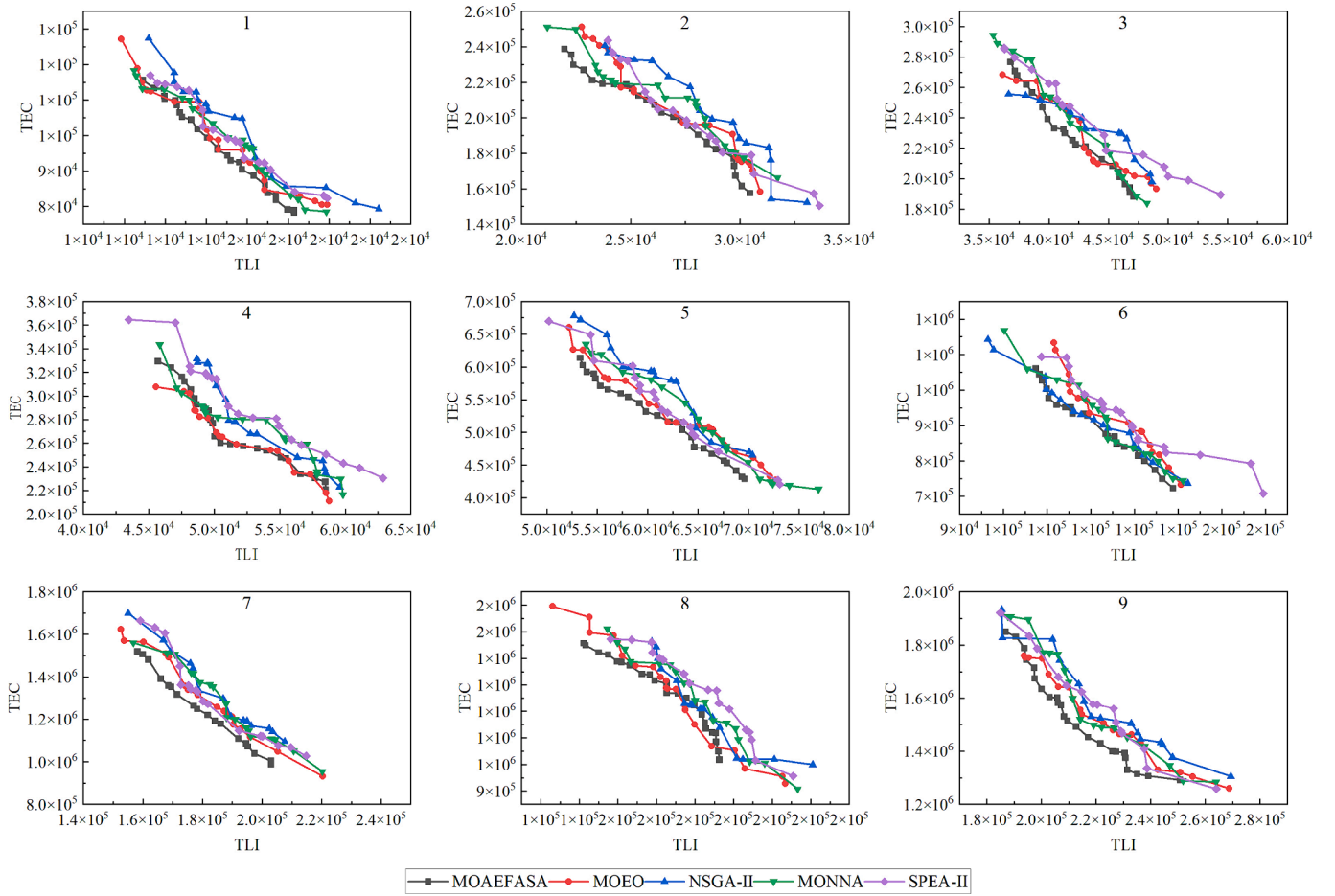


Fig. 10. Pareto fronts of meta-heuristic algorithm in different instances.

show better convergence and diversity of the Pareto sets obtained by MOAEFASA.

To further show the numerical distribution of ES and IGD, six instances are extracted forming the box plots. The results are shown in Fig. 12 and Fig. 13. It is obvious from the figures that MOAEFASA can achieve better results in ES and IGD in most instances, which indicates that the algorithm has stable solution performance.

In addition, the CPU runtimes of the five algorithms for solving different instances are compared. 6 different instances are selected and the average runtime for each instance is calculated. Fig. 14 shows the results of the runtimes. When the instance size is smaller than (60, 60), all five algorithms can obtain experimental results in a short time. However, when the instance size is larger than (60, 60), the runtimes of the algorithm grow significantly. For the largest problem (100, 100), MOAEFASA have the longest runtime of about 1300 s, which is still much smaller than the upper limit of 3600 s of Gurobi runtime set in the small-scale problems. Therefore, the runtimes of all five meta-heuristic algorithms are acceptable. In addition, MOAEFASA has the longest average runtime of all the instances. This is limited by the fact that the localized search strategy used by MOAEFASA takes some time. However, considering its outstanding performance on the solution quality proved by DPO, ES, and IGD, although a part of time efficiency is sacrificed as a cost, it can still be concluded that the MOAEFASA has a superior performance.

Finally, we first design a one-way ANOVA test for each evaluation metric at each scale problem, which is used to compare whether there are significant differences between multiple sets of experimental data. The significance level is 0.05. The results are shown in Table 9, which

shows that the p-values of all the evaluation metrics in all the scale experiments are less than 0.05. Further, the Turkey Test is added to better show the differences between each two arbitrary algorithms. The results are shown in Table 10 and Fig. 15. Table 10 shows the grouping of the five algorithms. It can be found that MOAEFASA is significantly different from the other four compared algorithms under the DPO and IGD metrics, while under the ES metric, MOAEFASA is significantly different from MOEO and NSGA-II. Fig. 15 shows the 95 % confidence intervals for the differences between each of the two algorithms under the Turkey test, which means that if the interval does not contain zeros, the corresponding means are significantly different. In summary, MOAEFASA has superior performance compared to the other four algorithms.

5.6. Effectiveness analysis of algorithm optimization strategies

The proposed algorithm introduces the diagonal linear random initialization strategy and the local optimization strategy based on the SARSA selection mechanism to optimize the algorithm. The diagonal linear random strategy aims to improve the diversity of the initial populations. The local optimization strategy based on the SARSA selection mechanism aims to improve the local search ability of the algorithm. To verify the effectiveness of the algorithm, the following experiments are conducted.

Firstly, in order to show the effectiveness of the local search strategy based on the SARSA selection mechanism, MOAEFASA is compared with DIMOAEFA, which does not consider the local search strategy, and basic MOAEFA. Nine instances are chosen in each of the three scales. The three algorithms are run independently 10 times. The average value of

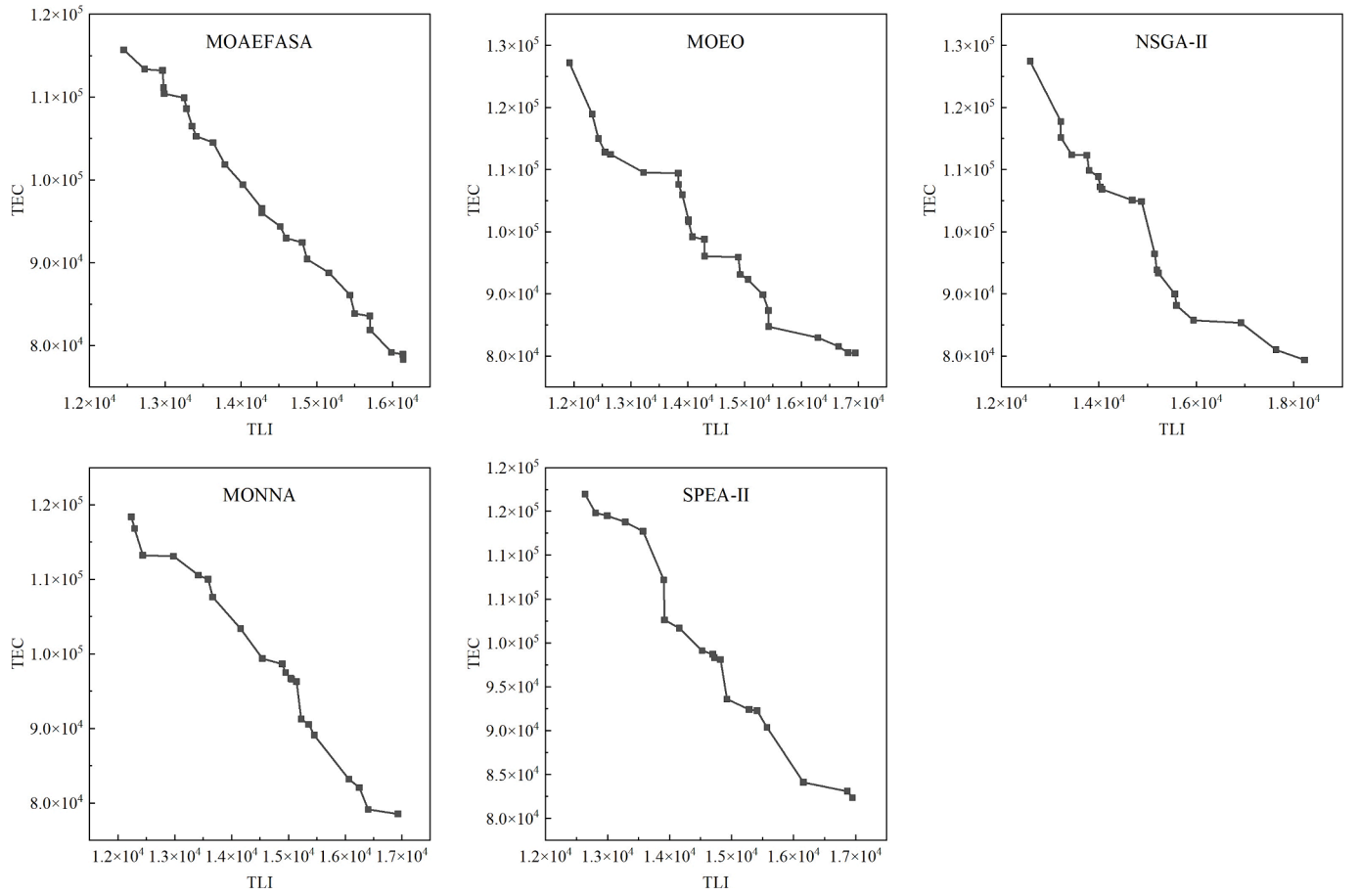


Fig. 11. Pareto fronts of meta-heuristic algorithms separately.

Table 6
Comparison of DPO metrics.

Group	T	S	MOAEFASA	MOEO	NSGA-II	MONNA	SPEA-II
Middle-scale	30	30	0.832	0.680	0.429	0.486	0.386
	30	40	0.802	0.676	0.302	0.724	0.304
	30	50	0.914	0.594	0.229	0.244	0.484
	40	30	0.978	0.618	0.306	0.737	0.503
	40	40	0.847	0.450	0.387	0.576	0.247
	40	50	0.673	0.749	0.220	0.388	0.485
	50	30	0.891	0.651	0.224	0.345	0.552
	50	40	0.690	0.442	0.144	0.750	0.222
	50	50	0.746	0.323	0.455	0.343	0.565
Large-scale	60	60	0.854	0.568	0.109	0.678	0.532
	60	80	0.966	0.554	0.139	0.246	0.436
	60	100	0.615	0.585	0.303	0.727	0.312
	80	60	0.971	0.601	0.327	0.204	0.292
	80	80	0.680	0.386	0.337	0.507	0.396
	80	100	0.755	0.720	0.365	0.200	0.208
	100	60	0.666	0.712	0.426	0.201	0.458
	100	80	0.875	0.713	0.143	0.722	0.551
	100	100	0.963	0.672	0.476	0.239	0.369
Ave.			0.818	0.594	0.296	0.462	0.406

the obtained DPO is used as the evaluation metrics. As shown in Fig. 16, the deviation of DPO values of the three algorithms is not large in small-scale instances, and all of them can produce good results. However, as the problem scale increases, the performance of DIMOAEFA and MOAEFA deteriorates seriously, while the performance of DIMOAEFA is usually better than that of MOAEFA. It shows that the local search strategy based on the SARSA selection mechanism can help the

algorithm find more Pareto solutions through multiple local searches, and effectively improve the solving accuracy of the algorithm. Meanwhile, Table 11 presents the ANOVA of the DPO values. The significance level is 0.05. As can be seen from the table, the p-value is greater than 0.05 in the small-scale instances. Therefore, in the small-scale instances, there is no obvious difference in the DPO values of the three algorithms. However, in medium-scale and large-scale instances, the p-values of the

Table 7
Comparison of ES metrics.

Group	T	S	MOAEFASA	MOEO	NSGA-II	MONNA	SPEA-II
Middle-scale	30	30	0.810	1.733	1.448	1.065	1.149
	30	40	0.772	1.780	1.330	0.981	0.928
	30	50	0.797	1.434	1.499	1.090	1.123
	40	30	0.928	1.485	1.545	1.121	0.992
	40	40	1.031	1.873	1.148	0.992	0.928
	40	50	1.008	1.478	1.357	1.088	0.935
	50	30	1.089	1.780	1.322	1.095	1.122
	50	40	0.980	1.405	1.382	1.139	0.932
	50	50	1.096	1.126	1.235	1.008	1.209
Large-scale	60	60	0.953	1.334	1.390	1.234	0.992
	60	80	1.087	1.938	1.269	1.057	1.140
	60	100	1.071	1.927	1.093	1.210	1.195
	80	60	0.978	1.286	1.654	1.139	0.939
	80	80	0.811	1.023	1.081	1.193	1.231
	80	100	0.978	1.849	1.271	0.958	0.924
	100	60	0.713	1.031	1.588	1.135	1.094
	100	80	0.980	1.642	1.482	1.000	0.973
	100	100	0.883	1.161	1.206	0.965	1.204
Ave.			0.943	1.516	1.350	1.082	1.056

Table 8
Comparison of IGD metrics.

Group	T	S	MOAEFASA	MOEO	NSGA-II	MONNA	SPEA-II
Middle-scale	30	30	0.357	0.281	0.807	0.308	0.529
	30	40	0.228	0.271	0.765	0.484	0.517
	30	50	0.296	0.413	0.607	0.532	0.794
	40	30	0.3	0.516	0.781	0.331	0.602
	40	40	0.382	0.406	0.637	0.382	0.606
	40	50	0.151	0.357	0.818	0.503	0.749
	50	30	0.283	0.372	0.568	0.501	0.605
	50	40	0.265	0.265	0.527	0.57	0.576
	50	50	0.414	0.521	0.687	0.396	0.748
Large-scale	60	60	0.192	0.35	0.544	0.505	0.669
	60	80	0.116	0.362	0.863	0.371	0.522
	60	100	0.262	0.543	0.844	0.453	0.534
	80	60	0.413	0.303	0.77	0.303	0.5
	80	80	0.324	0.277	0.778	0.391	0.648
	80	100	0.214	0.567	0.62	0.367	0.525
	100	60	0.193	0.33	0.588	0.572	0.537
	100	80	0.393	0.337	0.854	0.494	0.785
	100	100	0.167	0.297	0.621	0.328	0.556
Ave.			0.275	0.376	0.704	0.433	0.611

three algorithms are all less than 0.05, indicating that the performance of MOAEFASA is significantly more advantageous.

Then, the MOAEFASA is compared with LMOAEFA, using local search strategy, and basic MOAEFA to verify the effect of the diagonal linear random strategy on the performance of the algorithm. Similarly, nine instances are chosen at each of the three scales and each of the three algorithms run 10 times independently. The average value of the obtained ES and IGD is used as the evaluation metric. As can be seen from Fig. 17 and Fig. 18, the ES and IGD values of MOAEFASA are usually the smallest in the three scales of experiments, and the advantage is more obvious in large-scale instances. It shows that the algorithm can obtain higher quality Pareto solutions by increasing the diversity of the initial population. Finally, the ANOVA of ES and IGD values is conducted in Table 12. The significance level is 0.05. It can be seen from the table that all p-values are less than 0.05 under the three scales, indicating that MOAEFASA has obvious advantages over the two comparison algorithms in terms of uniformity, convergence and distribution.

5.7. Example application

5.7.1. Development of scheduling plans to guide actual production

To illustrate the applicability of the proposed model and methodology in real-world situations, we solve a test problem and give the final scheduling plan in this section. In the test problem, the factory has a production order that needs to be completed in 30 beats. The production order is shown in Table 13. In the production order, T represents the production cycle, and A, B and C represent the assembly types that need to be completed in the current production cycle.

This assembly work needs to be done on 30 workstations. Table 14 gives the parts that need to be assembled on different workstations for different vehicles and the Id of the parts are indicated in the grids below Type. The R on the left side of the table indicates the route to which the workstation s belongs. The rest of the parameters are set as shown in Section 5.1.

After calculation, the following solution to the problem is given. The Kanban quantity and material bin capacity corresponding to each part is shown in Table 15. The material distribution scheduling program is shown in Table 16. In the table, r represents the material distribution route, T represents the time for AGVs to distribute the materials, and the

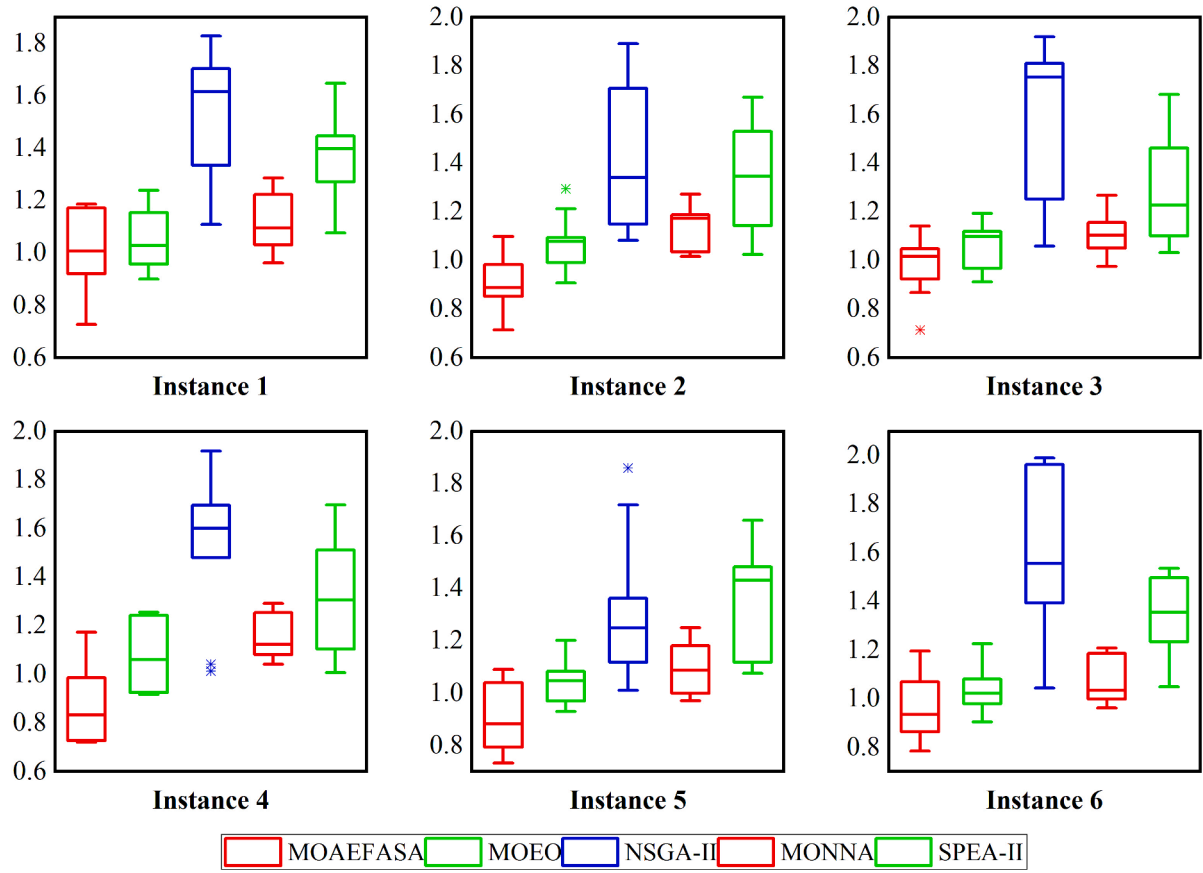


Fig. 12. Boxplots of ES.

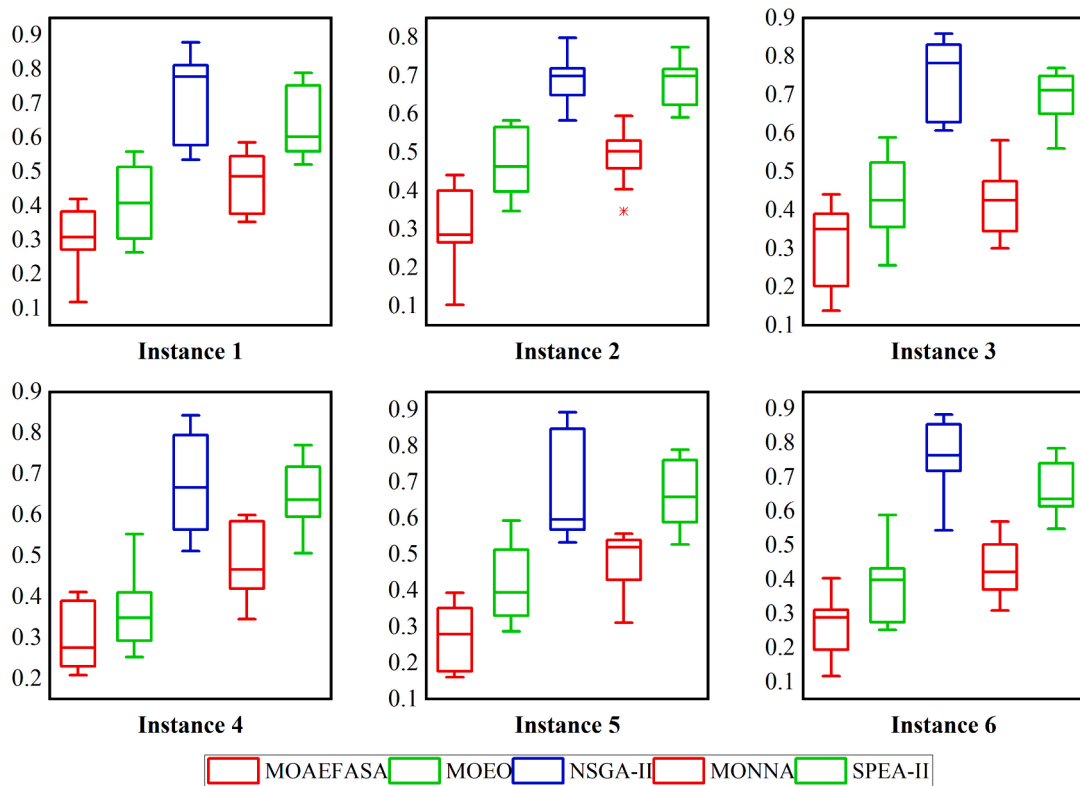


Fig. 13. Boxplots of IGD.

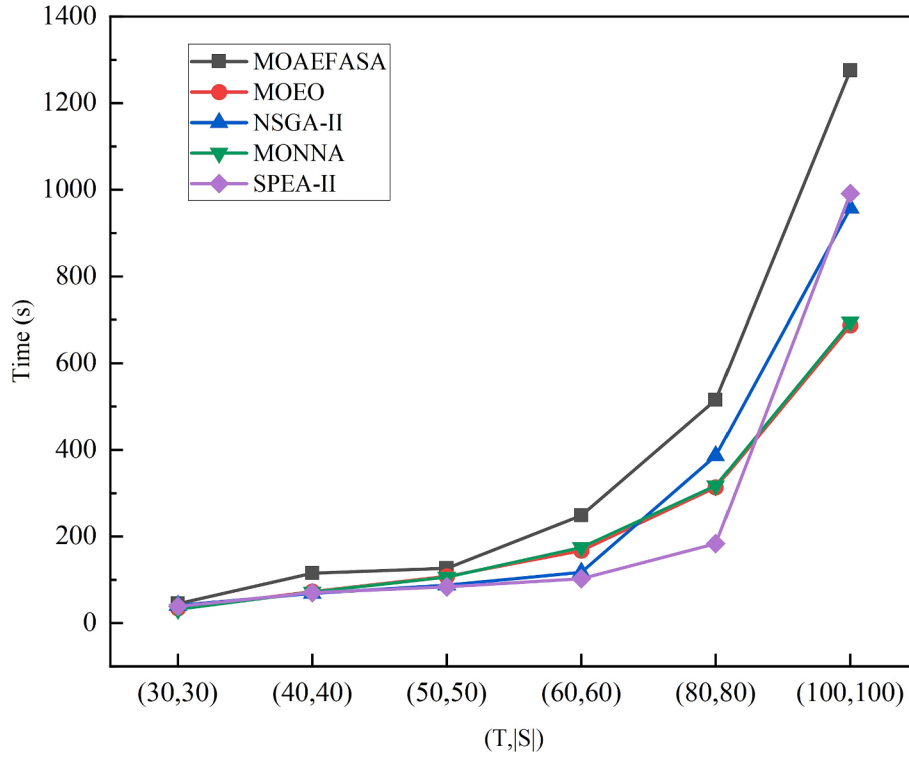


Fig. 14. The runtime of five meta-heuristic algorithms for different instances.

Table 9
ANOVA for DPO, ES, IGD at different scale problems.

Problem scales	Source	DPO				ES				IGD			
		DF	Adj SS	Adj MS	P	DF	Adj SS	Adj MS	P	DF	Adj SS	Adj MS	P
Medium	Algorithm	4	1.3678	0.34195	0	4	2.4351	0.60876	0	4	1.0122	0.253058	0
	Error	40	0.768	0.0192		40	0.8463	0.02116		40	0.3744	0.00936	
	Total	44	2.1358			44	3.2814			44	1.3866		
Large	Algorithm	4	1.5684	0.3921	0	4	1.632	0.40801	0	4	1.2152	0.3038	0
	Error	40	0.9577	0.02394		40	1.805	0.04512		40	0.44	0.011	
	Total	44	2.5261			44	3.437			44	1.6552		

Table 10
Turkey test results for DPO, ES, and IGD at all scale problems.

	Algorithm	N	Mean	Grouping			
DPO	MOAEFASA	18	0.8177	A			
	MOEO	18	0.5941	B			
	MONNA	18	0.4621	B			
	SPEA-II	18	0.4057	C			
	NSGA-II	18	0.2956	D			
ES	MOEO	18	1.5158	A			
	NSGA-II	18	1.35	A			
	MONNA	18	1.0817	B			
	SPEA-II	18	1.0561	B			
	MOAEFASA	18	0.9425	B			
IGD	NSGA-II	18	0.7044	A			
	SPEA-II	18	0.6112	B			
	MONNA	18	0.4328	C			
	MOEO	18	0.376	C			
	MOAEFASA	18	0.275	D			

corresponding grid denotes the parts to be distributed. Finally, the corresponding values for this solution on the TLI and TEC objectives are 14,841 and 117580, respectively.

5.7.2. Validate the effectiveness of optimize the Kanban quantity and material bin capacity

In order to help the actual production management, the effectiveness for the collaborative optimization of Kanban quantity and material bin capacity is verified. A total of 10 instances, $T = 80, S = \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$, are selected. According to 5.1 and theorem 2, the theoretical lower bound of Kanban quantity Nk_i^{min} for each part can be calculated. Then, the material bin capacity of each part needs to be the maximum value Cb_i^{max} . Therefore, the actual inventory capacity Iv_i^s of each part at the assembly lines should meet $Iv_i^s \geq Nk_i^{min} \hat{A} \cdot Cb_i^{max}$. In each instance, two real-world scenarios are set up. In the first scenario, the Kanban quantity of each part $NK_i = Nk_i^{min}$ and the material bin capacity is a random integer in the interval $[Cb_i^{max}, 2Cb_i^{max}]$. In the second scenario, the material bin capacity of each part is $Cb_i = \lfloor Iv_i^s / Nk_i^{min} \rfloor$, and the Kanban quantity is a random integer in $[NK_i^{max}, 2NK_i^{max}]$. Each instance is run 10 times using MOAEFASA. The two objectives of the Pareto solutions are averaged. The results are compared with the TLI and TEC values calculated by the two actual scenarios. The results are shown in Fig. 19.

The GAP value calculation method is shown in Equation (53).

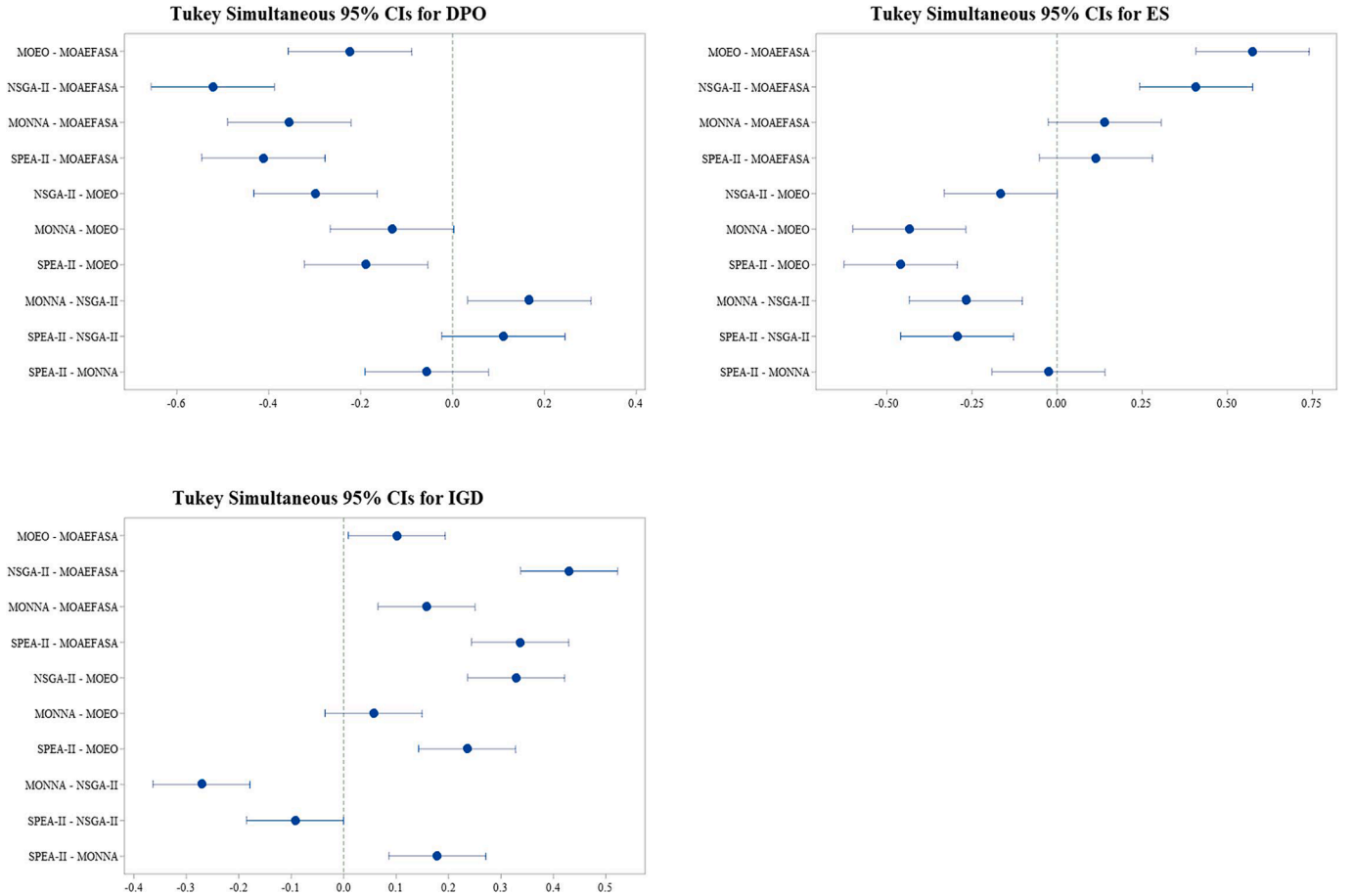


Fig. 15. Turkey test results for DPO, ES and IGD at all scale problems.

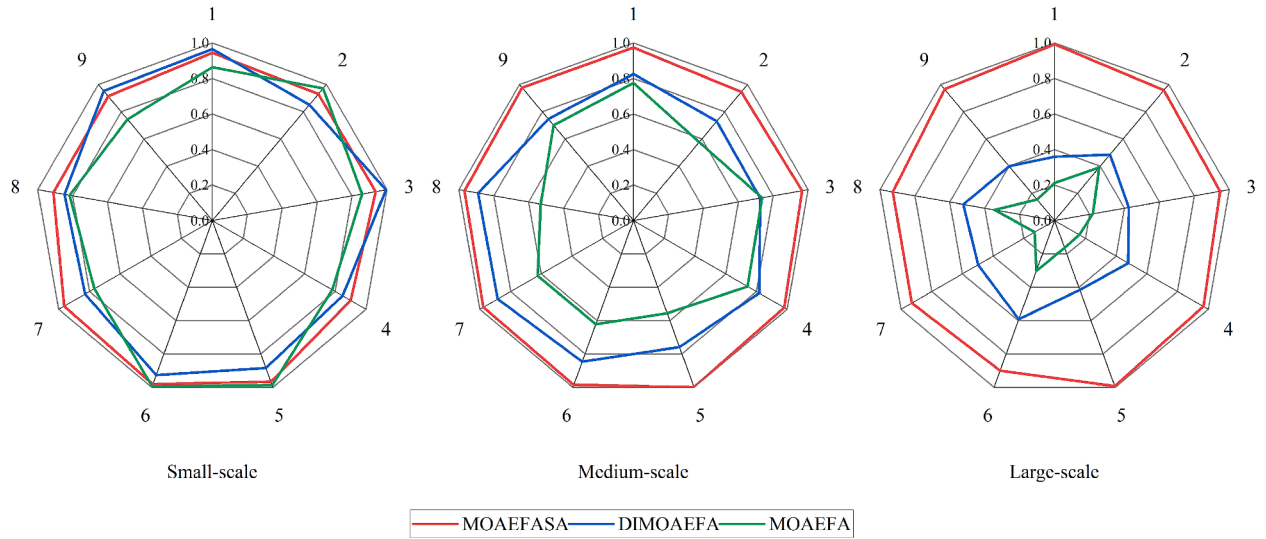


Fig. 16. Radar maps of DPO.

$$Gap = \min(Obj_{Sc1}, Obj_{Sc2}) - Obj_{Al} / Obj_{Al} \times 100\% \quad (50)$$

Where, Obj_{Sc1} and Obj_{Sc2} represent the objective values obtained in scenarios 1 and 2, respectively, and Obj_{Al} represents the average objective value obtained by the algorithm.

The experimental results show that on the premise of ensuring no shortages on the assembly lines, compared with the two scenarios of

only considering the Kanban quantity or the material bin capacity, the combination optimization of these two can effectively reduce the TLI and the TEC simultaneously. In all instances, the minimum GAP for TLI is 14.2 % and the maximum GAP is 30 %, while the minimum GAP for TEC is 10.1 % and the maximum GAP is 36.2 %. As a result, when compared to optimizing only the Kanban quantity or the material bin capacity, the combination optimization of the Kanban quantity and the

Table 11
ANOVA for DPO.

Source	Small				Medium				Large			
	DF	Adj SS	Adj MS	P	DF	Adj SS	Adj MS	P	DF	Adj SS	Adj MS	P
Algorithm	2	0.025	0.012	0.094	2	0.471	0.236	0	2	2.450	1.225	0
Error	24	0.113	0.005		24	0.099	0.004		24	0.118	0.005	
Total	26	0.137			26	0.570			26	2.568		

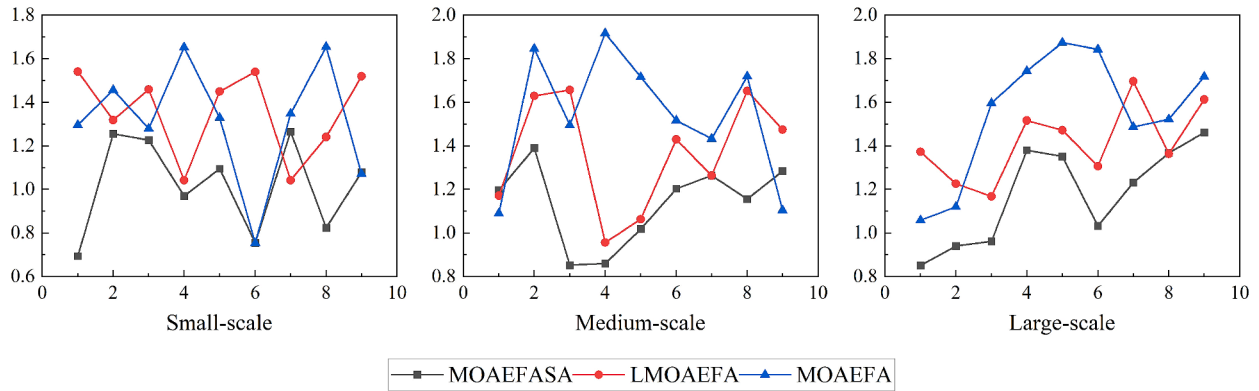


Fig. 17. Line charts of ES.

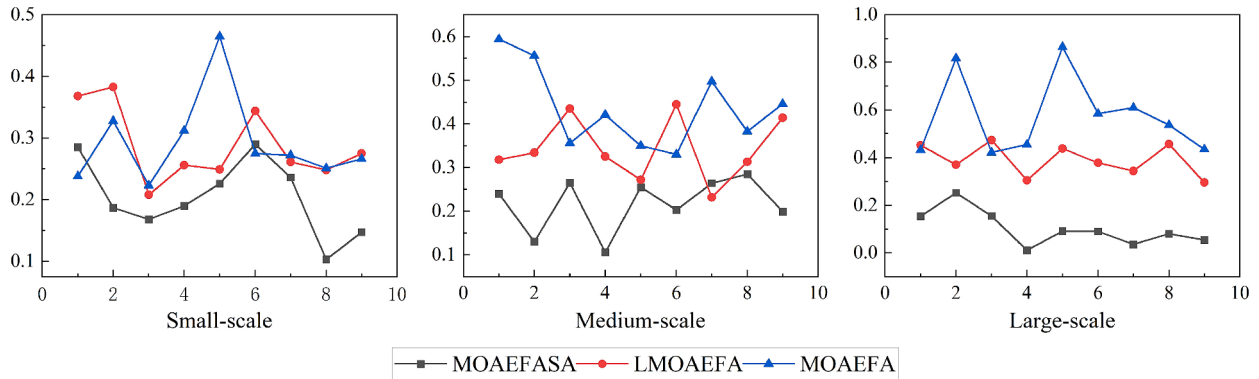


Fig. 18. Line charts of IGD.

Table 12
ANOVA for ES, IGD.

Source		Small				Medium				Large			
		DF	Adj SS	Adj MS	P	DF	Adj SS	Adj MS	P	DF	Adj SS	Adj MS	P
ES	Algorithm	2	0.600	0.300	0.012	2	0.730	0.365	0.01	2	0.653	0.327	0.009
	Error	24	1.337	0.056		24	1.545	0.064		24	1.355	0.056	
	Total	26	1.937			26	2.275			26	2.009		
IGD	Algorithm	2	0.045	0.023	0.013	2	0.221	0.110	0	2	1.013	0.507	0
	Error	24	0.103	0.004		24	0.146	0.006		24	0.303	0.013	
	Total	26	0.148			26	0.367			26	1.316		

Table 13
Production orders for 30 production cycles.

T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Production order	B	C	B	A	A	A	B	B	B	A	B	C	B	C	A
T	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Production order	C	B	C	B	A	C	A	A	C	C	C	A	B	A	A

Table 14

Parts to be assembled at each workstation for different vehicles.

R	S	Type A	Type B	Type C	R	S	Type A	Type B	Type C
1	1	2, 3	1, 2	1, 2	4	16	49	48, 49, 50	48, 50
	2	4, 5	4, 5	5	17	52, 53	51, 54	53, 54	
	3	7, 8	6, 7	9	18	55, 57	55, 56, 57	55	
	4	10, 11	10	10	19	60	58, 59, 60	59, 60	
	5	13, 14, 15	12, 14, 15	12, 13, 15	20	61, 62, 63	63	62, 63	
2	6	17	16	16, 17	5	21	65, 66, 67	66	64, 66
	7	18, 19, 20	18, 19	19, 20, 21	22	69	68	70	
	8	23, 25	22, 23, 25	22, 24, 25	23	74	72, 73, 74	71, 72, 74	
	9	26	26, 28	27	24	75, 77, 78	75, 76, 77	75, 76, 77	
	10	31	30	29, 30, 31	25	79	80	80	
3	11	32, 33, 34	32	32	6	26	82, 83	83	81
	12	35, 36, 37	35, 36, 38	35, 38	27	84	84	84, 85, 86	
	13	41	39, 40	40	28	87, 88, 89	87, 89	87	
	14	42	42, 43	42	29	93	90, 91, 92	90	
	15	44, 45, 47	44	44, 45, 46	30	96	95, 97	94	

material bin capacity is extremely important.

6. Conclusion

In this paper, we take the energy consumption objective as one of the factors to consider in the formulation of a milk-run material distribution scheduling scheme, and propose a mathematical model for the MMDSPCE. In addition, a multi-objective optimization algorithm, MOAEFASA, is used to solve the MMDSPCE. The effectiveness and superiority of the algorithm are verified by experiments. This paper can be summarized in the following four points.

- (1) A bi-objective material distribution scheduling problem, referred to as MMDSPCE, is introduced, which focuses on achieving a

balance between the line-side inventory index and the energy consumption index within a milk-run material distribution system. Building upon this concept, the study incorporates considerations for Kanban quantity and material bin capacity.

- (2) A mathematical model for the optimization of the proposed MMDSPCE is established. Moreover, the accuracy of the model is demonstrated with the help of the Epsilon constraint method.
- (3) To solve the problem, a multi-objective optimization algorithm, MOAEFASA, is proposed. The algorithm mixed AEFA and NSGA-II as the overall framework to improve the performance of the algorithm. Then, several optimization strategies are proposed to optimize the algorithm. The diagonal linear random initialization strategy is used to generate the initial populations, which improves the diversity and uniformity of the initial populations. The SARSA-based selection mechanism is used to obtain the optimal local optimization scheme.
- (4) MOAEFASA is compared with MOEO, NSGA-II, MONNA and SPEA-II. Experimental results show the superiority of the algorithm. Meanwhile, the effectiveness of the proposed optimization strategies is also illustrated. Finally, through solving a specific instance and analyzing the impact of collaborative optimization of Kanban quantity and material bin capacity, the solution to the problems for actual production management are provided.

In future research, the real-time scheduling problem should be considered. The problems studied in this paper assume that the demand is fixed and there is no perturbation from unknown factors. However, the actual production needs to prevent the assembly line from being shut down. Therefore, in the JIT system, it is a great challenge to react quickly to unknown disturbances without affecting production. In addition, other indicators of the milk-run material distribution system can be optimized. In future research, models targeting line-side inventory minimization, energy consumption minimization, vehicle utilization maximization, vehicle number minimization, or other indicators can be developed. Lastly, given the intricacy of the problem, it is worthwhile to delve deeper into refining the algorithm's coding, aiming to devise a coding scheme better tailored to material distribution scheduling problems. This endeavor holds the potential to enhance the algorithm's efficiency and performance significantly when addressing such challenges.

CRedit authorship contribution statement

Binghai Zhou: Conceptualization, Formal analysis, Funding

Table 15

The Kanban quantity and material bin capacity for each part.

Id	NK _i	Cb _i	Id	NK _i	Cb _i	Id	NK _i	Cb _i	Id	NK	Cb _i	Id	NK _i	Cb _i
1	4	2	21	4	2	41	4	2	61	4	1	81	3	2
2	4	3	22	4	2	42	3	2	62	4	3	82	4	2
3	4	2	23	4	2	43	4	2	63	3	2	83	4	2
4	3	2	24	4	2	44	4	3	64	4	2	84	4	2
5	4	3	25	4	2	45	3	2	65	4	2	85	3	2
6	4	2	26	4	3	46	4	2	66	3	2	86	4	2
7	4	3	27	3	2	47	4	3	67	4	2	87	4	2
8	4	2	28	4	2	48	4	3	68	4	3	88	5	2
9	4	2	29	3	2	49	4	2	69	4	2	89	4	2
10	4	2	30	4	3	50	4	2	70	4	2	90	4	2
11	3	2	31	4	2	51	4	2	71	4	2	91	4	2
12	3	2	32	4	2	52	4	2	72	4	3	92	4	3
13	3	2	33	3	2	53	4	3	73	4	3	93	4	2
14	4	2	34	3	2	54	4	1	74	4	2	94	4	2
15	4	2	35	4	2	55	3	3	75	3	2	95	4	2
16	5	2	36	4	2	56	4	2	76	4	2	96	4	2
17	4	3	37	3	1	57	4	2	77	4	2	97	4	2
18	4	1	38	3	2	58	4	2	78	4	2			
19	4	3	39	4	2	59	3	2	79	4	3			
20	5	2	40	4	2	60	4	2	80	4	2			

Table 16
Material distribution scheduling program for 30 production cycles.

<i>T</i>	<i>r</i> = 1	<i>r</i> = 2	<i>r</i> = 3	<i>r</i> = 4	<i>r</i> = 5	<i>r</i> = 6
1			32,35,38,40,42	50,54,59,60,63	74,75,76,77,80	
2	2,4,5,6,10,15	16,19,22,23,25,30				83,84,89,90,91,97
3			32,35,36,37,42,44	54,55,56,57,58,61	65,66,72,74,75,77	
4	3,4,7,13,14,15	17,20,23,25,26,31				83,84,88,89,93,96
5			32,35,36,37,41,44	52,53,54,55,57,61	67,69,74,75,78,79	
6	2,7,8,11,12,13	16,18,22,23,25				83,84,87,89,90,92
7			32,35,36,37,38,40	50,54,56,57,58,61	68,73,74,75,76,80	
8	2,4,6,10,12,16	18,22,25,26				83,84,89,90,91,97
9			35,36,37,38,40,41	49,54,57,63	67,74,75,76,78,80	
10	1,3,5,7,8,15	18,20,23,26,30				82,87,91,95,97
11			36,38,42,45	51,55,56,58,60	64,65,66,74,77	
12	2,10,11,14	17,18,24,25,28				81,84,85,86,89,90
13			32,35,38,40,44,46	54,59,63	70,72,74,75,76,80	
14	1,4,12	16,18,23,25,27				87,90,93,95,96
15			39,42,43	49,54,60	70,71,73,76,77	
16	5,10,13,15	16,19,20,21,31				81,84,85,86,89,91
17			33,34,41	45,48,50,59,63	65,66,68,71,77	
18	1,2,4,7,14	17,18,26,29,30				84,87,88,92,97
19			32,38,42,44	54,55,59,60,63	69,74,75,76,80	
20	10,12,15	22,25,28				81,84,85,86,89,90
21			35,36,37,42	49,54,57,60,61,63	65,67,68,77	
22	1,5,13	18,19,24,31				83,87,88,96
23			32,40,45	50,54,59,60,61,63	64,66,72,75	
24	4,10,14,15	18,23,25				83,84,87,88,92
25			35,37,42,44	49,53,55,60,61,62	70,71,74,77	
26	9,12,13,15	20,22,27,29,31				81,82,84,86,94
27			32,38,42,44,45,46	48,50,54,59,60,63	66,75,80	
28	2,5,10,11,13,15	19,21,24,25,27,31				82,83,87,88,96
29			33,34,35,36,38,40	49,52,57,60,61,63	65,74,76,77	
30						

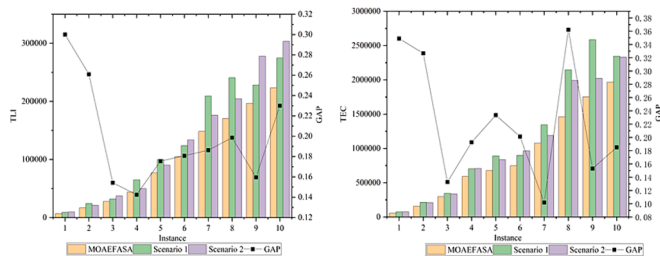


Fig. 19. Comparison of two objectives in three situations.

acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing. **Mingda Wen:** Methodology, Software, Validation, Visualization, Writing – original draft.

Data availability

Data will be made available on request.

References

Adegboye, O. R., & Deniz Ulker, E. (2023). Hybrid artificial electric field employing cuckoo search algorithm with refraction learning for engineering optimization problems. *Science Reports*, 13(1), 4098. <https://doi.org/10.1038/s41598-023-31081-1>

Akkad, M. Z., & Bányaí, T. (2023). Energy consumption optimization of milk-run-based in-plant supply solutions: an industry 4.0 approach. *Processes*, 11(3). <https://doi.org/10.3390/pr11030799>

Anita, & Yadav, A. (2019). AEFA: Artificial electric field algorithm for global optimization. *Swarm and Evolutionary Computation*, 48, 93–108. <https://doi.org/10.1016/j.swevo.2019.03.013>

Anita, Yadav, A., & Kumar, N. (2020). Artificial electric field algorithm for engineering optimization problems. *Expert Systems with Applications*, 149. <https://doi.org/10.1016/j.eswa.2020.113308>

Bányai, T. (2023). Energy efficiency of AGV-drone joint in-plant supply of production lines. *Energies*, 16(10). <https://doi.org/10.3390/en16104109>

Barak, S., Moghdani, R., & Maghsoudlou, H. (2021). Energy-efficient multi-objective flexible manufacturing scheduling. *Journal of Cleaner Production*, 283. <https://doi.org/10.1016/j.jclepro.2020.124610>

Bi, J., Zhou, Y. Q., Tang, Z. H., & Luo, Q. F. (2022). Artificial electric field algorithm with inertia and repulsion for spherical minimum spanning tree. *Applied Intelligence*, 52 (1), 195–214. <https://doi.org/10.1007/s10489-021-02415-1>

Boysen, N., & Emde, S. (2014). Scheduling the part supply of mixed-model assembly lines in line-integrated supermarkets. *European Journal of Operational Research*, 239(3), 820–829. <https://doi.org/10.1016/j.ejor.2014.05.029>

Briand, C., He, Y., & Nguereu, S. U. (2018). Energy-efficient planning for supplying assembly lines with vehicles. *Euro Journal on Transportation and Logistics*, 7(4), 387–414. <https://doi.org/10.1007/s13676-018-0129-8>

Cao, M. T., Hoang, N. D., Nhu, V. H., & Bui, D. T. (2022). An advanced meta-learner based on artificial electric field algorithm optimized stacking ensemble techniques for enhancing prediction accuracy of soil shear strength. *Engineering with Computers*, 38(3), 2185–2207. <https://doi.org/10.1007/s00366-020-01116-6>

Ciemnoczolowski, D. D., & Bozer, Y. A. (2013). Performance evaluation of small-batch container delivery systems used in lean manufacturing – Part 2: number of Kanban and workstation starvation. *International Journal of Production Research*, 51(2), 568–581. <https://doi.org/10.1080/00207543.2012.656331>

, 2006Coello, C.A.C. (2006). Evolutionary multi-objective optimization: A historical view of the field. *Ieee Computational Intelligence Magazine*, 1(1), 28–36. <https://doi.org/10.1109/Mci.2006.1597059>

Díaz, C. J. J., & Ocampo-Martínez, C. (2019). Energy efficiency in discrete-manufacturing systems: Insights, trends, and control strategies. *Journal of Manufacturing Systems*, 52, 131–145. <https://doi.org/10.1016/j.jmsy.2019.05.002>

Domingo, R., Alvarez, R., Peña, M. M., & Calvo, R. (2007). Materials flow improvement in a lean assembly line: a case study. *Assembly Automation*, 27(2), 141–147. <https://doi.org/10.1108/01445150710733379>

Elmi, Z., Li, B. K., Liang, B. B., Lau, Y. Y., Borowska-Stefanska, M., Wisniewski, S., & Dulebenets, M. A. (2023). An epsilon-constraint-based exact multi-objective optimization approach for the ship schedule recovery problem in liner shipping. *Computers & Industrial Engineering*, 183. <https://doi.org/10.1016/j.cie.2023.109472>

Emde, S., Abedinnia, H., & Glock, C. H. (2018). Scheduling electric vehicles making milk-runs for just-in-time delivery. *IIE Transactions*, 50(11), 1013–1025. <https://doi.org/10.1080/04725854.2018.1479899>

Emde, S., & Schneider, M. (2018). Just-in-time vehicle routing for in-house part feeding to assembly lines. *Transportation Science*, 52(3), 657–672. <https://doi.org/10.1287/trsc.2018.0824>

Facchini, F., Mossa, G., Sassanelli, C., & Digiesi, S. (2023). IoT-based milk-run routing for manufacturing system: an application case in an automotive company. *International Journal of Production Research*, 1–20. <https://doi.org/10.1080/00207543.2023.2254408>

Faccio, M., Gamberi, M., & Persona, A. (2013). Kanban number optimisation in a supermarket warehouse feeding a mixed-model assembly system. *International Journal of Production Research*, 51(10), 2997–3017. <https://doi.org/10.1080/00207543.2012.751516>

- Fathi, M., Rodríguez, V., Fontes, D. B. M. M., & Alvarez, M. J. (2015). A modified particle swarm optimisation algorithm to solve the part feeding problem at assembly lines. *International Journal of Production Research*, 54(3), 878–893. <https://doi.org/10.1080/00207543.2015.1090032>
- Francuz, Á., & Bánya, T. (2022). Optimisation of milkrun routes in manufacturing systems in the automotive industry. *Cutting & Tools in Technological System*, 96, 32–41. <https://doi.org/10.20998/2078-7405.2022.96.04>
- Hanson, R., & Finnsgård, C. (2014). Impact of unit load size on in-plant materials supply efficiency. *International Journal of Production Economics*, 147, 46–52. <https://doi.org/10.1016/j.ijpe.2012.08.010>
- Hou, Y. S., Wang, H. F., Fu, Y. P., Gao, K. Z., & Zhang, H. (2023). Multi-Objective brain storm optimization for integrated scheduling of distributed flow shop and distribution with maximal processing quality and minimal total weighted earliness and tardiness. *Computers & Industrial Engineering*, 179. <https://doi.org/10.1016/j.cie.2023.109217>
- Izci, D., Ekinci, S., Orenc, S., & Demiroren, A. (2020). Improved artificial electric field algorithm using nelder-mead simplex method for optimization problems. In: 2020 4th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT) (pp. 1–5): IEEE.
- Kilic, H. S., & Durmusoglu, M. B. (2013). A mathematical model and a heuristic approach for periodic material delivery in lean production environment. *International Journal of Advanced Manufacturing Technology*, 69(5–8), 977–992. <https://doi.org/10.1007/s00170-013-5082-y>
- Kilic, H. S., Durmusoglu, M. B., & Baskak, M. (2012). Classification and modeling for in-plant milk-run distribution systems. *International Journal of Advanced Manufacturing Technology*, 62(9–12), 1135–1146. <https://doi.org/10.1007/s00170-011-3875-4>
- Li, Q., Bai, Y. G., & Gao, W. F. (2021). Improved initialization method for metaheuristic algorithms: a novel search space view. *IEEE Access*, 9, 121366–121384. <https://doi.org/10.1109/Access.2021.3073480>
- Liu, Q., Cui, C., & Fan, Q. (2022). Self-adaptive constrained multi-objective differential evolution algorithm based on the state-action-reward-state-action method. *Mathematics*, 10(5). <https://doi.org/10.3390/math10050813>
- Mehouachi, I., Trojette, M., & Grayaa, K. (2023). MONNA: Multi-objective neural network algorithm for the optimal location of electric vehicle charging infrastructure in Tunis city. *Journal of Cleaner Production*, 431. <https://doi.org/10.1016/j.jclepro.2023.139837>
- Miqueo, A., Gracia-Cadarso, M., Torralba, M., Gil-Vilda, F., & Yagüe-Fabra, J. A. (2023). Multi-model in-plant logistics using milkruns for flexible assembly systems under disturbances. *An Industry Study Case. Machines*, 11(1). <https://doi.org/10.3390/machines11010066>
- Naderipour, A., Abdul-Malek, Z., Bin Mustafa, M. W., & Guerrero, J. M. (2021). A multi-objective artificial electric field optimization algorithm for allocation of wind turbines in distribution systems. *Applied Soft Computing*, 105. <https://doi.org/10.1016/j.asoc.2021.107278>
- Niroomand, S. (2021). Hybrid artificial electric field algorithm for assembly line balancing problem with equipment model selection possibility. *Knowledge-Based Systems*, 219. <https://doi.org/10.1016/j.knosys.2021.106905>
- Peng, T., & Zhou, B. H. (2019). Hybrid bi-objective gray wolf optimization algorithm for a truck scheduling problem in the automotive industry. *Applied Soft Computing*, 81. <https://doi.org/10.1016/j.asoc.2019.105513>
- Petwal, H., & Rani, R. (2020). An improved artificial electric field algorithm for multi-objective optimization. *Processes*, 8(5). <https://doi.org/10.3390/pr8050584>
- Premkumar, M., Jangir, P., Sowmya, R., Alhelou, H. H., Mirjalili, S., & Kumar, B. S. (2022). Multi-objective equilibrium optimizer: framework and development for solving multi-objective optimization problems. *Journal of Computational Design and Engineering*, 9(1), 24–50. <https://doi.org/10.1093/jcde/qwab065>
- Rauf, H. T., Shoaib, U., Lali, M. I., Alhaisoni, M., Irfan, M. N., & Khan, M. A. (2020). Particle swarm optimization with probability sequence for global optimization. *IEEE Access*, 8, 110535–110549. <https://doi.org/10.1109/Access.2020.3002725>
- Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems (Vol. 37): Citeseer.
- Sipahioglu, A., & Altin, I. (2019). A mathematical model for in-plant Milk-Run routing. *Pamukkale University Journal of Engineering Sciences-Pamukkale Üniversitesi Muhendislik Bilimleri Dergisi*, 25(9), 1050–1055. <https://doi.org/10.5505/pajes.2019.03366>
- Tasoglu, G., & Ali Ilgin, M. (2023). A simulation-based genetic algorithm approach for the simultaneous consideration of reverse logistics network design and disassembly line balancing with sequencing. *Computers & Industrial Engineering*. <https://doi.org/10.1016/j.cie.2023.109794>
- Vahedi-Nouri, B., Arbabi, H., Jolai, F., Tavakkoli-Moghaddam, R., & Bozorgi-Amiri, A. (2022). Bi-objective collaborative electric vehicle routing problem: mathematical modeling and matheuristic approach. *Journal of Ambient Intelligence and Humanized Computing*. <https://doi.org/10.1007/s12652-021-03689-6>
- Wang, C., Yang, C., & Zhang, T. (2023). Order planning with an outsourcing strategy for a make-to-order/make-to-stock production system using particle swarm optimization with a self-adaptive genetic operator. *Computers & Industrial Engineering*, 182. <https://doi.org/10.1016/j.cie.2023.109420>
- Wolpert, D. H., & Macready, W. G. (1997). No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1), 67–82. <https://doi.org/10.1109/4235.585893>
- Yang, Y. K., Liu, J. C., Tan, S. B., & Wang, H. H. (2019). A multi-objective differential evolutionary algorithm for constrained multi-objective optimization problems with low feasible ratio. *Applied Soft Computing*, 80, 42–56. <https://doi.org/10.1016/j.asoc.2019.02.041>
- Zhang, S. Z., Lee, C. K. M., Choy, K. L., Ho, W., & Ip, W. H. (2014). Design and development of a hybrid artificial bee colony algorithm for the environmental vehicle routing problem. *Transportation Research Part D-Transport and Environment*, 31, 85–99. <https://doi.org/10.1016/j.trd.2014.05.015>
- Zhou, B., & He, Z. (2021). A novel hybrid-load AGV for JIT-based sustainable material handling scheduling with time window in mixed-model assembly line. *International Journal of Production Research*, 61(3), 796–817. <https://doi.org/10.1080/00207543.2021.2017056>
- Zhou, B. H., & Peng, T. (2019). Scheduling just-in-time part replenishment of the automobile assembly line with unrelated parallel machines. *Proceedings of the Institution of Mechanical Engineers Part C-Journal of Mechanical Engineering Science*, 233(14), 5113–5130. <https://doi.org/10.1177/0954406219839083>
- Zhou, B. H., & Shen, C. Y. (2018). Multi-objective optimization of material delivery for mixed model assembly lines with energy consideration. *Journal of Cleaner Production*, 192, 293–305. <https://doi.org/10.1016/j.jclepro.2018.04.251>
- Zhou, B. H., & Zhao, L. W. (2022). A multi-objective decomposition evolutionary algorithm based on the double-faced mirror boundary for a milk-run material feeding scheduling optimization problem. *Computers & Industrial Engineering*, 171. <https://doi.org/10.1016/j.cie.2022.108385>
- Zhou, B. H., & Zhao, Z. (2023). An adaptive artificial bee colony algorithm enhanced by Deep Q-Learning for milk-run vehicle scheduling problem based on supply hub. *Knowledge-Based Systems*, 264. <https://doi.org/10.1016/j.knosys.2023.110367>
- Zhou, B. H., & Zhu, Z. X. (2021). Multi-objective optimization of green scheduling problems of part feeding for mixed model assembly lines based on the robotic mobile fulfillment system. *Neural Computing & Applications*, 33(16), 9913–9937. <https://doi.org/10.1007/s00521-021-05761-w>